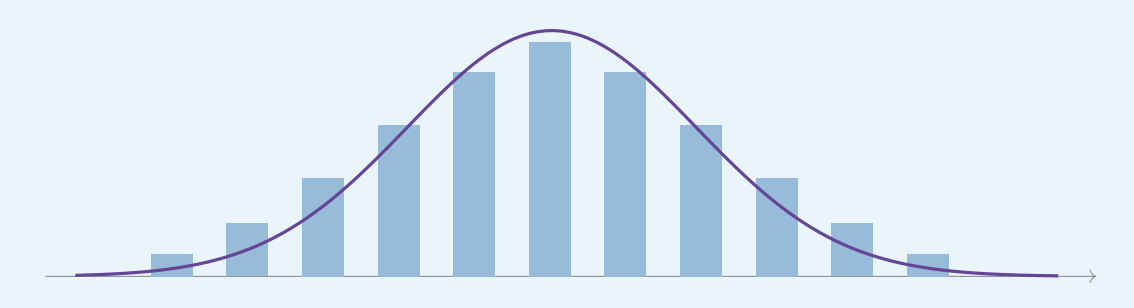
PROBABILITY

Complete Learning Notes

From events to distributions, dependence, and limits

Version V4.1 October 19, 2026

Organized by Shang Wu



Definition: The learning route

Model the experiment → specify support → select a representation → derive the calculation → verify the result.

Understanding before memorization. Detailed derivations, worked examples, support diagrams, standard solution workflows, and a mind map after every chapter.

Course scope follows the supplied syllabus image. Reference: Sheldon Ross, *A First Course in Probability*, ninth edition, Chapters 1–8. Text, examples, questions, solutions, and diagrams are newly prepared; no school-specific past examination was supplied.

Course Scope and Coverage Map

The six lines of the supplied syllabus are expanded into twelve learning chapters. The textbook is Ross, ninth edition (2014); its printed page 1 is PDF page 14 in the supplied file. Section references below use the book's printed numbering, not PDF offsets.

Syllabus requirement	Learning chapters and skills	Ross reference / practice IDs
Spaces, axioms, conditioning, independence	1–2: counting, event algebra, probability, Bayes, conditional independence	Chs. 1–3; P1.1–P2.6
Bernoulli experiments; discrete, continuous, mixed variables	3–5: CDF, atoms, support and principal distribution models	4.1–4.10; 5.1–5.6; P3.1–P5.6
Distribution, density and probability functions	3–5, 7–8: PMF/PDF/CDF, transformations, convolution and joint laws	4.10; 5.7; 6.1–6.7; P3.1–P8.6
Expectation, variance, moments, transformations, Jensen, vectors	6–9: moment existence, indicators, inequalities, Jacobians, conditional moments, covariance	4.3–4.5; 6.7; 7.2–7.6; 8.5; P6.1–P9.6
Joint and conditional laws, correlation, Gaussian vectors	8–10: dependence, conditional expectation, linear maps, normal conditioning	Ch. 6; 7.4–7.6; 7.8; P8.1–P10.6
Functions, Chebyshev, LLN, CLT	11–12: PGF, MGF, characteristic functions; bounds and normal/Poisson approximation	7.7; 8.2–8.6; P11.1–P12.6

Error Check: What the source does and does not establish

The phrase "Typical functions" in the syllabus is ambiguous. To avoid a gap, Chapter 11 includes probability-generating, moment-generating, and characteristic functions. Characteristic functions and their use in a CLT proof are an explicit supplement to Ross's MGF-centered treatment. Multivariate Gaussian details also expand the concise textbook discussion to match the vector requirement. No claim about the school's favorite examination topics is made without actual past papers.

Prerequisites and notation

$\mathbb{P}$ denotes probability, $\mathbb{E}$ expectation, Var variance, and Cov covariance. $F_X(x) = \mathbb{P}(X \le x)$, p_X is a mass function, and f_X a density. Φ and ϕ are the standard normal CDF and density. Vectors are columns, and covariance matrices use transpose, not componentwise squaring. Integrals over unspecified limits mean the full support. A distribution's parameters are always stated: exponential and gamma parameters are *rates*; geometric variables count trials unless specified otherwise.

Calculus prerequisites are differentiation, elementary integrals, integration by parts, convergent series, double integrals over a drawn region, and a Jacobian determinant. Linear algebra prerequisites are matrix multiplication, quadratic forms, orthogonal eigenvectors, positive definiteness, and diagonalization.

Standard Workflow: The six-line answer pattern

1. Define the experiment, variables, parameters, and support.
2. Translate the event into inequalities, including endpoint conventions.
3. Name the method and verify independence, integrability, or regularity assumptions.

4. Show the decisive counting, sum, integral, derivative, or matrix calculation.
5. Report the requested probability, law, moment, bound, or approximation.
6. Check normalization, dimensions, support, limiting cases, and whether the answer is exact.

Priority without unsupported predictions

High-dependency skills are support analysis, conditioning, indicator sums, transformations, covariance, Gaussian standardization, and selecting exact laws versus limit approximations. These are priorities because they connect multiple syllabus topics, not because examination frequencies were measured. Chapters end with method maps, while the separate practice volume offers six multipart problems per chapter in different testing directions. Advanced Markov chains, entropy, coding, and simulation from Ross Chapters 9–10 are outside the supplied syllabus and are not silently treated as compulsory.

Contents

Course Scope and Coverage Map i

1 Experiments, Events, and Counting 1
- 1.1 A model comes before a fraction . . . 1
- 1.2 Event algebra and consequences of the axioms . . . 1
- 1.3 Counting with a consistent sample space . . . 2
- 1.4 Continuity of probability . . . 2
- 1.5 Chapter Mind Map . . . 3

2 Conditional Probability and Independence 3
- 2.1 Conditioning changes the reference population . . . 3
- 2.2 Total probability and Bayes from disjoint pieces . . . 3
- 2.3 Pairwise, mutual, and conditional independence . . . 4
- 2.4 Sequential trials and first-step analysis . . . 5
- 2.5 Chapter Mind Map . . . 5

3 Random Variables, CDFs, and Mixed Laws 5
- 3.1 A random variable is a function . . . 5
- 3.2 Mass functions, densities, and atoms . . . 6
- 3.3 Mixed laws require two pieces . . . 6
- 3.4 Quantiles and inverse-CDF construction . . . 7
- 3.5 Chapter Mind Map . . . 7

4 Bernoulli Experiments and Discrete Distributions 8
- 4.1 Bernoulli and binomial: fixed numbers of trials . . . 8
- 4.2 Geometric and negative binomial: waiting times . . . 8
- 4.3 Hypergeometric: finite sampling without replacement . . . 9
- 4.4 Poisson and multinomial models . . . 9

4.5 Chapter Mind Map 10

5 Principal Continuous Distributions 10

5.1 Uniform models and geometric probability 10
5.2 Exponential survival and its memoryless property 11
5.3 Normal distribution: normalization and standardization 11
5.4 Gamma, beta, and related families 12
5.5 Heavy tails and transformed positive laws 12
5.6 Chapter Mind Map 13

6 Expectation, Moments, and Inequalities 13

6.1 Expectation is a weighted integral, not an outcome prediction 13
6.2 Indicators and tail sums 14
6.3 Markov, Chebyshev, and one-sided bounds 14
6.4 Jensen and Cauchy–Schwarz with their assumptions 14
6.5 Chapter Mind Map 15

7 Transformations, Convolution, and Order Statistics 15

7.1 The CDF method works beyond monotonicity 15
7.2 Independent sums and convolution 16
7.3 Joint transformations: inverse map and Jacobian 16
7.4 Minima, maxima, and ranks 17
7.5 Chapter Mind Map 18

8 Joint Laws and Conditional Expectation 18

8.1 Joint CDFs and marginals 18
8.2 Conditional distributions are normalized slices 19
8.3 The tower rule and total variance 19
8.4 Prediction as an orthogonal decomposition 20
8.5 Chapter Mind Map 20

9 Covariance, Correlation, and Random Vectors 20

9.1 Cross terms carry dependence 20
9.2 Correlation and its exact equality cases 21
9.3 Covariance matrices and linear maps 21
9.4 Best linear prediction and random sums 22
9.5 Chapter Mind Map 22

10 Gaussian Vectors and Conditional Normal Laws 23

10.1 Construction, density, and singular cases 23
10.2 Linear combinations and orthogonal normal coordinates 24
10.3 Conditional normal law: derive via an independent residual 24
10.4 Quadratic forms and chi-square laws 25

10.5 Chapter Mind Map 25

11 PGFs, MGFs, and Characteristic Functions **25**

11.1 Three transforms with different domains 25
11.2 MGFs derive moments and identify laws 26
11.3 Characteristic functions: always available 26
11.4 Random sums and Poisson splitting 27
11.5 Chapter Mind Map 27

12 Inequalities, Laws of Large Numbers, and the CLT **28**

12.1 What kind of convergence is being claimed? 28
12.2 Weak LLN: concentration of the sample mean 28
12.3 Central limit theorem: fluctuations on a shrinking scale 29
12.4 Discrete continuity correction and approximation selection 29
12.5 Exponential bounds and failure checks 30
12.6 Chapter Mind Map 30

A Formula Index and Method Selection **31**

A.1 Core distribution index 31
A.2 Short transform index 31
A.3 Six recurring decisions 31
A.4 Source and completeness record 31
A.5 Frequently Needed Integrals and Series 32
A.6 Stable Calculations and Finite Uniform Laws 32

1 Experiments, Events, and Counting

Source map: Ross Chapters 1–2; practice P1.1–P1.6.

1.1 A model comes before a fraction

A probability problem has two layers: a set of possible outcomes and a rule assigning probabilities to events. A die total is not as informative as the ordered pair of die faces; the totals are not equally likely even when all ordered pairs are. Choosing a convenient outcome space is part of the solution, not a cosmetic first line.

Definition: Probability space

A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$. The sample space Ω contains outcomes. The event family $\mathcal{F}$ is a sigma-algebra: it contains Ω, complements, and countable unions. The measure satisfies $\mathbb{P}(A) \geq 0$, $\mathbb{P}(\Omega) = 1$, and

$$\mathbb{P}\left(\bigcup_{n\geq 1} A_n\right) = \sum_{n\geq 1} \mathbb{P}(A_n) \quad \text{for pairwise disjoint } A_n.$$

For finite spaces one usually takes every subset as an event. The sigma-algebra matters in infinite spaces, but no construction of nonmeasurable sets is required here.

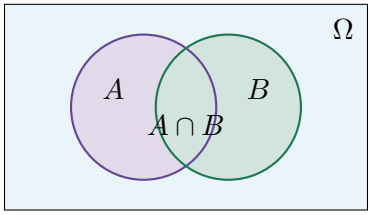


Figure 1.1: Overlapping events must not be counted twice. The surrounding rectangle is the entire sample space.

1.2 Event algebra and consequences of the axioms

The complement A^c means "not A," the union $A \cup B$ means "at least one," and the intersection $A \cap B$ means "both." De Morgan's rules are $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

Since $\Omega = A \,\dot\cup\, A^c$, additivity gives $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. If $A \subseteq B$, write $B = A \,\dot\cup\, (B \setminus A)$, obtaining monotonicity and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$. Finally decompose $A \cup B$ into three disjoint pieces. The intersection occurs twice in $\mathbb{P}(A) + \mathbb{P}(B)$, hence

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

For three events, add single probabilities, subtract the three pair intersections, and add the triple intersection. For n events, inclusion–exclusion alternates over nonempty subsets. The union bound follows either by discarding overlaps through disjointification or by the pointwise inequality $\mathbf{1}_{\{\cup_i A_i\}} \leq \sum_i \mathbf{1}_{\{A_i\}}$:

$$\mathbb{P}(\cup_i A_i) \leq \sum_i \mathbb{P}(A_i).$$

Worked Example: Recovering a complete event table

Suppose $\mathbb{P}(A) = .55$, $\mathbb{P}(B) = .40$, and $\mathbb{P}(A \cup B) = .75$. The overlap is $.55 + .40 - .75 = .20$. Therefore the probabilities of $A \cap B$, $A \cap B^c$, $A^c \cap B$, and $A^c \cap B^c$ are $.20, .35, .20, .25$. They are nonnegative and add to one. This four-cell table is often more reliable than several separate formulas.

1.3 Counting with a consistent sample space

For equiprobable finite outcomes only, $\mathbb{P}(A) = |A|/|\Omega|$. The multiplication principle gives $n_1 \cdots n_r$ possibilities when each stage has the stated number of choices conditional on previous choices. Ordered draws without replacement give $n!/(n-r)!$; unordered subsets give $\binom{n}{r}$ because each subset has $r!$ orderings. Repeated indistinguishable types give $n!/(n_1! \cdots n_k!)$ arrangements.

For nonnegative integers $x_1 + \cdots + x_k = n$, encode n stars and $k-1$ separators: $\binom{n+k-1}{k-1}$ solutions. For positive integers replace x_i by $1 + y_i$, giving $\binom{n-1}{k-1}$ when $n \geq k$. These counts do not mean that occupancy vectors are equally probable when labeled balls independently enter boxes.

Worked Example: Sampling a committee

Choose four people uniformly from six engineers and five chemists. The probability of exactly two of each is

$$\frac{\binom{6}{2}\binom{5}{2}}{\binom{11}{4}} = \frac{150}{330} = \frac{5}{11}.$$

The denominator counts unordered groups, and the numerator does too. If a specified pair of engineers may not serve together, subtract $\binom{5}{2} = 10$ favorable groups containing both; the restricted event with exactly two of each has probability $140/330 = 14/33$ under the original unrestricted sampling experiment. If sampling itself is restricted, the denominator must also change.

Worked Example: No repeated labels and the complement method

Draw five labels independently and uniformly from twenty types. The probability of no repeat is $(20 \cdot 19 \cdot 18 \cdot 17 \cdot 16)/20^5$. Thus at least one repeat has probability one minus this fraction. Counting "a repeated pair" directly double-counts outcomes with several collisions. A product naturally computes the complement.

1.4 Continuity of probability

If $A_n \uparrow A$, define disjoint increments $D_1 = A_1$, $D_n = A_n \setminus A_{n-1}$. Countable additivity gives $\mathbb{P}(A) = \sum_n \mathbb{P}(D_n) = \lim_n \mathbb{P}(A_n)$. If $A_n \downarrow A$, apply the increasing result to complements to obtain the same limit. These statements justify taking limits of events, rather than treating infinity as a finite counting problem.

Worked Example: An outcome of probability zero can occur

For U uniform on $[0,1]$, intervals $A_n = [0, 1/n]$ decrease to $\{0\}$. Their probabilities are $1/n$, so $\mathbb{P}(U = 0) = 0$. Nevertheless 0 is an allowed outcome. Countably many probability-zero points have zero union probability; an uncountable union need not, as $[0,1]$ demonstrates.

Standard Workflow: Event and counting questions

Define the experiment; decide order and replacement; justify equal likelihood; translate the event into disjoint cases or a complement; count numerator and denominator consistently; check bounds and extreme parameters. For infinite limits, name the increasing or decreasing event sequence.

Error Check: Disjointness is not independence

Disjoint events cannot both happen. Independence says information about one does not change the other's probability. Two disjoint events with positive probabilities cannot be independent because their intersection has probability zero but the product of their probabilities is positive.

1.5 Chapter Mind Map

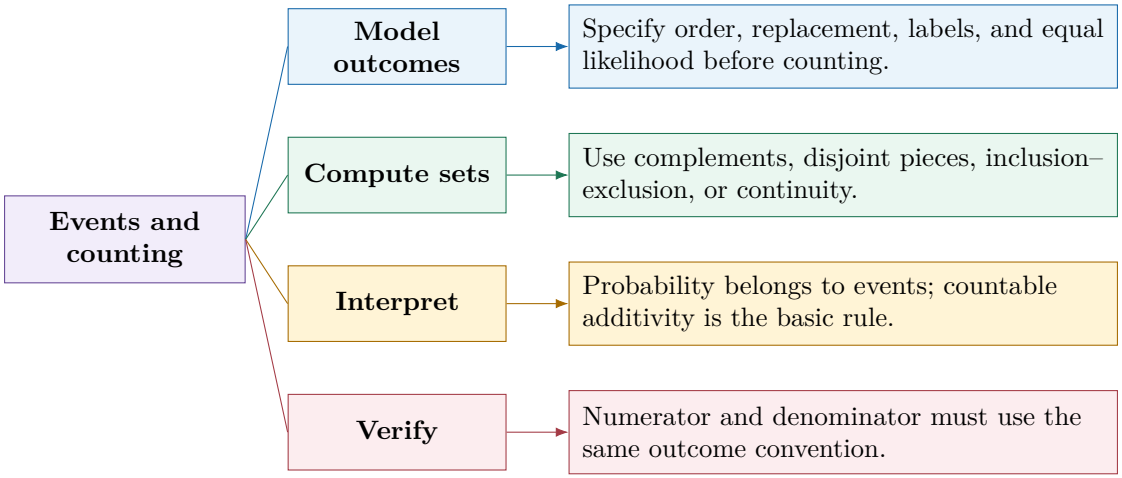


Figure 1.2: The chapter's model, calculation route, interpretation, and checks.

2 Conditional Probability and Independence

Source map: Ross 3.2–3.5; practice P2.1–P2.6.

2.1 Conditioning changes the reference population

Definition: Conditional probability and multiplication

If $\mathbb{P}(B) > 0$, then $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$. Consequently $\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B)$. For several events, repeated multiplication gives

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1) \prod_{j=2}^{n} \mathbb{P}(A_j \mid A_1 \cap \cdots \cap A_{j-1}),$$

provided conditioning events have positive probability. Zero-probability branches contribute zero and need not be assigned an arbitrary quotient.

For fixed B, the map $A \mapsto \mathbb{P}(A \mid B)$ is itself a probability measure: it is nonnegative, assigns one to Ω, and preserves countable disjoint sums after intersection with B. Thus all event identities remain valid within a conditioned population.

Worked Example: Condition on information, not a story shortcut

Two fair dice are rolled. Given that at least one die is six, the reference set has $36 - 25 = 11$ ordered pairs. Only $(6, 6)$ has both dice six, so the answer is $1/11$. Given instead that the first die is six, there are six pairs and the answer is $1/6$. The wording identifies different conditioning events.

2.2 Total probability and Bayes from disjoint pieces

If (H_i) partitions Ω, then $A = \dot{\cup}_i (A \cap H_i)$, so

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid H_i)\mathbb{P}(H_i).$$

Apply the conditional definition to $H_j \cap A$ and insert that sum in the denominator:

Key Result: Bayes's formula

$$\mathbb{P}(H_j \mid A) = \frac{\mathbb{P}(A \mid H_j)\mathbb{P}(H_j)}{\sum_i \mathbb{P}(A \mid H_i)\mathbb{P}(H_i)}, \qquad \mathbb{P}(A) > 0.$$

Posterior odds equal prior odds times the likelihood ratio:

$$\frac{\mathbb{P}(H \mid A)}{\mathbb{P}(H^c \mid A)} = \frac{\mathbb{P}(H)}{\mathbb{P}(H^c)} \frac{\mathbb{P}(A \mid H)}{\mathbb{P}(A \mid H^c)}$$

when the displayed denominators are positive.

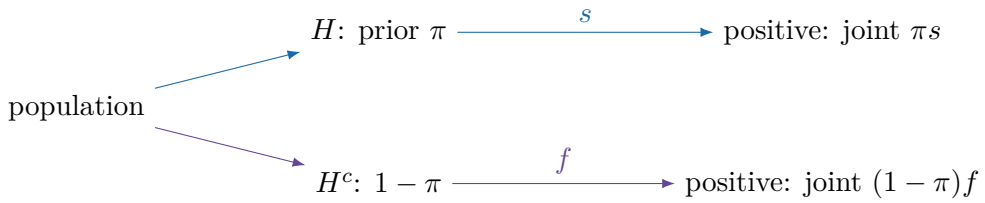


Figure 2.1: A positive observation may arise from either hypothesis. Normalize the desired branch by the sum of all positive branches.

Worked Example: A low base rate and repeated evidence

An item is faulty with prior probability .02. A test is positive with probability .95 if faulty and .04 otherwise. Then

$$\mathbb{P}(+) = .02(.95) + .98(.04) = .0582, \quad \mathbb{P}(H \mid +) = .019/.0582 = 95/291 \approx .3265.$$

For two tests that are independent *conditional on fault status*, the positive-positive branch has probability $.02(.95)^2 + .98(.04)^2 = .019618$. Its faulty part is .01805, giving posterior $9025/9809 \approx .9201$. The tests are not unconditionally independent: the latent fault status links them.

2.3 Pairwise, mutual, and conditional independence

A and B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, a definition valid even for zero-probability events. It implies independence of complements: subtract the intersection from $\mathbb{P}(A)$ to get $\mathbb{P}(A \cap B^c) = \mathbb{P}(A)(1 - \mathbb{P}(B))$.

A collection is mutually independent if every finite subcollection has joint probability equal to the product. Pairwise tests alone omit higher-order interactions. Conditional independence given C means $\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\mathbb{P}(B \mid C)$, not unconditional independence.

Worked Example: Pairwise independent but not mutually independent

Toss two fair bits U, V. Let $A = \{U = 1\}$, $B = \{V = 1\}$, and $C = \{U = V\}$. Each has probability $1/2$, and each pair intersection has probability $1/4$. But $A \cap B \cap C = \{(1, 1)\}$ has probability $1/4$, not $1/8$. The third event is determined by the first two bits.

Worked Example: Conditioning can create dependence

Independent fair bits satisfy $\mathbb{P}(U = 1, V = 1) = 1/4$. Condition on $U + V = 1$. Now $U = 1$ forces $V = 0$, so the conditional joint probability of two ones is zero, while both conditional marginal probabilities are $1/2$. Independence is not automatically preserved when a common event is imposed.

2.4 Sequential trials and first-step analysis

For independent repeated trials, disjoint outcomes A, B on each trial have probabilities $a, b > 0$. Let r be the chance A occurs before B. Conditioning on the first trial gives $r = a + (1 - a - b)r$, so $r = a/(a + b)$. This recurrence uses a genuine restart: after neither event, the future experiment is unchanged. The same reasoning fails when trials have memory.

Standard Workflow: A standard conditioning solution

Write the hidden categories and their prior probabilities; draw a tree if helpful; compute branch probabilities by multiplication; sum branches for the evidence; divide to obtain the posterior. For an independence question, specify whether the test is pairwise, mutual, or conditional and check the corresponding products.

Check / common mistake. Do not interchange $\mathbb{P}(A \mid B)$ and $\mathbb{P}(B \mid A)$. Do not multiply two likelihoods unless their conditional independence is part of the model.

2.5 Chapter Mind Map

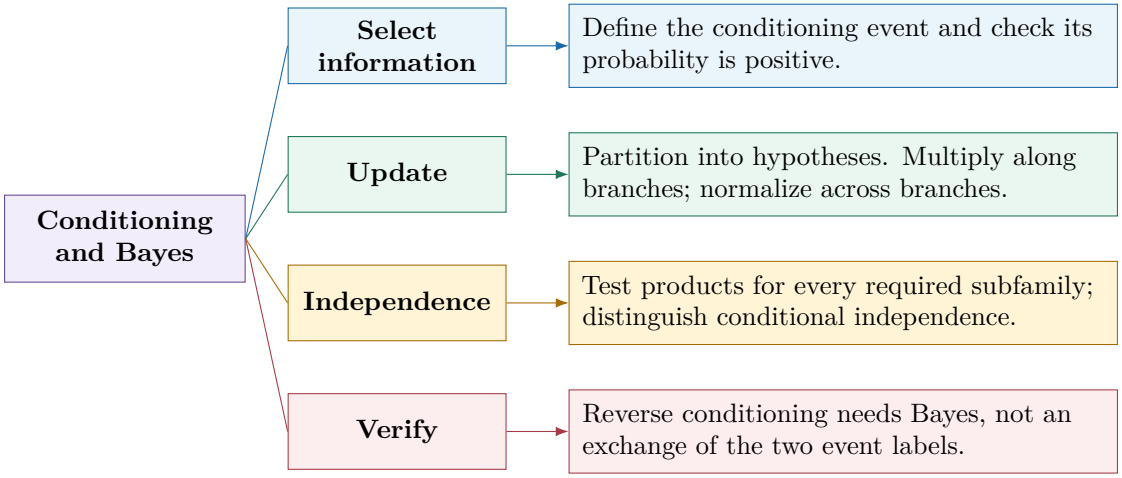


Figure 2.2: The chapter's model, calculation route, interpretation, and checks.

3 Random Variables, CDFs, and Mixed Laws

Source map: Ross 4.1–4.2, 4.10, 5.1, 7.9; mixed-law details explicitly expanded; P3.1–P3.6.

3.1 A random variable is a function

A real random variable $X : \Omega \to \mathbb{R}$ assigns a number to each outcome, with measurable inverse images. Its distribution is the probability rule induced on the real line. Distinct variables can have the same distribution without being equal on each outcome. A Bernoulli bit and its complement have the same law at parameter $1/2$, but their sum is exactly one, not a sum of independent bits.

Definition: The universal representation: the CDF

$F_X(x) = \mathbb{P}(X \le x)$. Every CDF is nondecreasing, right-continuous, and has limits zero at $-\infty$ and one at $+\infty$. Write $F_X(a-) = \lim_{x \uparrow a} F_X(x)$. Then

$$\mathbb{P}(X = a) = F_X(a) - F_X(a-),$$
$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a),$$

$$\mathbb{P}(a \le X \le b) = F_X(b) - F_X(a-).$$

Monotonicity follows from nested events. Right continuity follows from $\{X \le x + 1/n\} \downarrow \{X \le x\}$. The two tail limits follow from continuity of probability on expanding or shrinking half-lines. The jump formula follows by subtracting $\{X < a\}$ from $\{X \le a\}$.

3.2 Mass functions, densities, and atoms

For a discrete variable, $p_X(x) = \mathbb{P}(X = x)$ on a finite or countable support, with $p_X \ge 0$ and $\sum_x p_X(x) = 1$. Then $F_X(t) = \sum_{x \le t} p_X(x)$. An absolutely continuous variable has $F_X(t) = \int_{-\infty}^{t} f_X(x)\mathrm{d}x$ with $f_X \ge 0$ and integral one. Its point probabilities vanish. Where differentiable, $F_X' = f_X$, but differentiating alone loses any jumps.

Error Check: Continuous CDF does not always mean a density exists

In this course, "continuous random variable" usually means absolutely continuous. Mathematically a singular continuous CDF is also possible; it has no jumps but need not equal the integral of its ordinary derivative. The atomic-plus-density formulas below cover mixed laws of the stated kind, not every conceivable distribution.

Worked Example: Normalize a mass function and reconstruct the CDF

Let $p_X(k) = c(k+1)$ for $k = 0, 1, 2$. Since $c(1+2+3) = 1$, $c = 1/6$. The CDF is zero below zero, $1/6$ on $[0, 1)$, $1/2$ on $[1, 2)$, and one on $[2, \infty)$. Thus $\mathbb{P}(1 \le X < 2) = 1/3$, not $F(2) - F(1) = 1/2$; the latter is the probability of $1 < X \le 2$.

3.3 Mixed laws require two pieces

Suppose X has atoms a_j with masses w_j and a continuous component $h \ge 0$ satisfying $\sum_j w_j + \int h = 1$. Then

$$F_X(t) = \sum_{a_j \le t} w_j + \int_{-\infty}^{t} h(x)\mathrm{d}x, \quad \mathbb{E}[g(X)] = \sum_j w_j g(a_j) + \int g(x)h(x)\mathrm{d}x$$

whenever the expectation is defined. The function h integrates to the continuous probability, not necessarily to one.

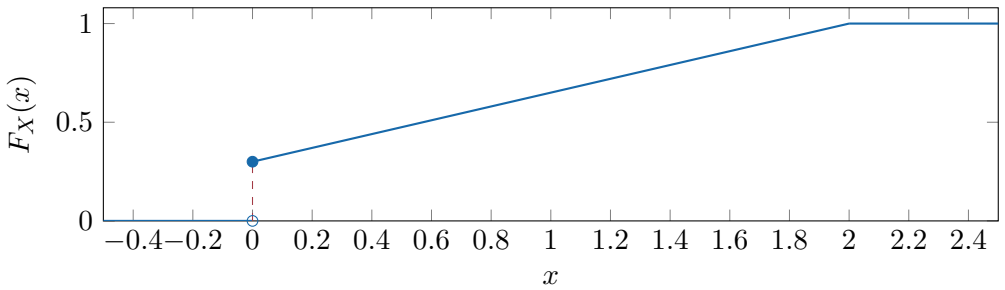


Figure 3.1: An atom of size 0.3 at zero and a continuous component of density 0.35 on $(0, 2)$ produce a jump followed by a linear CDF.

Worked Example: An atom plus a uniform component

Let $X = 0$ with probability .3; otherwise draw uniformly on $(0, 2)$. Then $F(x) = 0$ for $x < 0$, $F(x) = .3 + .35x$ for $0 \le x < 2$, and $F(x) = 1$ for $x \ge 2$. The continuous density is .35, not .5. Thus $\mathbb{P}(X \le .4) = .44$, $\mathbb{E}X = .7$, $\mathbb{E}X^2 = .35(8/3) = 14/15$, and $\operatorname{Var} X = 14/15 - .49 = 133/300$.

Worked Example: Censoring creates a point mass

For $T \sim \text{Exp}(\lambda)$ and fixed $c > 0$, set $Y = \min(T, c)$. For $0 \le y < c$, $F_Y(y) = 1 - \mathrm{e}^{-\lambda y}$; at c it jumps to one. The atom is $\mathbb{P}(Y = c) = \mathrm{e}^{-\lambda c}$. The density $\lambda \mathrm{e}^{-\lambda y}$ applies only on $(0, c)$ and integrates to $1 - \mathrm{e}^{-\lambda c}$. Replacing the atom by a density endpoint value would lose positive probability.

3.4 Quantiles and inverse-CDF construction

For $0 < u < 1$, define $Q(u) = \inf\{x : F_X(x) \ge u\}$. If $U \sim \text{Unif}(0, 1)$, then $Q(U)$ has CDF F_X: the event $Q(U) \le x$ agrees up to null boundary issues with $U \le F_X(x)$. This generalized inverse handles jumps and flat CDF portions.

Worked Example: Quantiles with a jump

For the atom-plus-uniform example, $Q(u) = 0$ for $0 < u \le .3$, while $Q(u) = (u - .3)/.35$ for $.3 < u < 1$. The median is $4/7$. An ordinary inverse applied to the linear portion would produce negative values for $u < .3$, contradicting the support.

Standard Workflow: A law specified by a piecewise CDF

Check monotonicity and right continuity; find all jumps using left limits; differentiate only smooth intervals; check atomic mass plus continuous integral is one; calculate requested events with endpoint-aware CDF differences; include both parts in expectations.

3.5 Chapter Mind Map

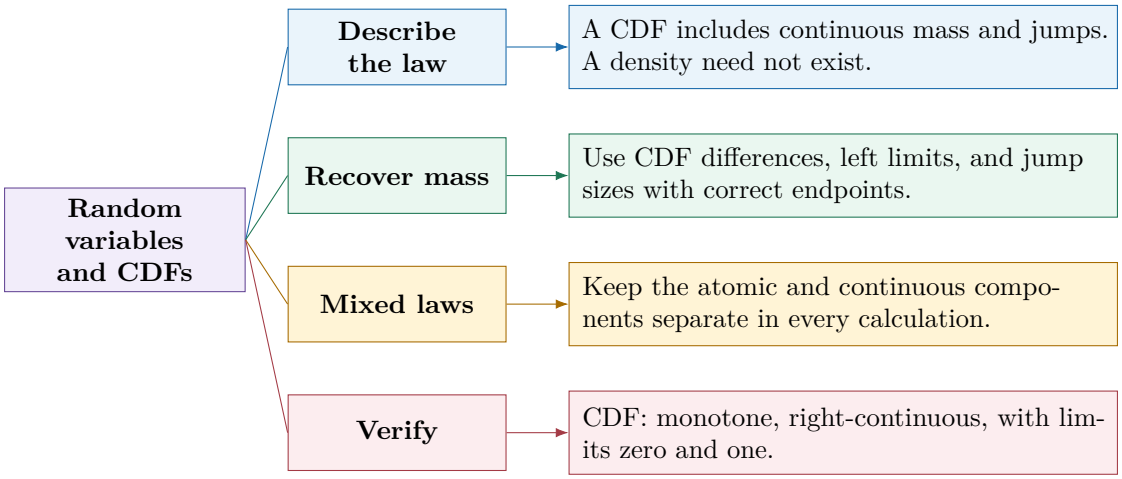


Figure 3.2: The chapter's model, calculation route, interpretation, and checks.

4 Bernoulli Experiments and Discrete Distributions

Source map: Ross 4.6–4.9, 6.1; P4.1–P4.6.

4.1 Bernoulli and binomial: fixed numbers of trials

If I indicates success, $\mathbb{P}(I=1)=p$, $\mathbb{E}I=p$, and $I^2=I$ implies $\operatorname{Var} I=p(1-p)$. For independent indicators $I_1,\dots,I_n$ with a common success probability, $X=\sum I_i$ is binomial. Each fixed sequence with k successes has probability $p^k(1-p)^{n-k}$; there are $\binom{n}{k}$ such sequences. Therefore

$$\mathbb{P}(X=k)=\binom{n}{k}p^k(1-p)^{n-k},\quad k=0,\dots,n.$$

Linearity gives $\mathbb{E}X=np$; independence makes all cross covariances zero, so $\operatorname{Var} X=np(1-p)$. If success probabilities differ, the sum remains a sum of Bernoulli variables but is generally not binomial.

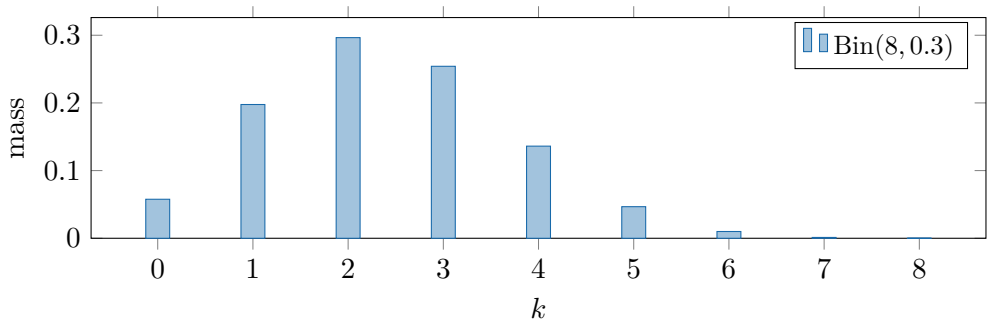


Figure 4.1: A binomial law assigns mass to integer counts only; a smooth curve is not its exact mass function.

Worked Example: A reliability threshold

Five independent modules work with probability .8 each. At least four must work. Then $\mathbb{P}(X\ge 4)=\binom{5}{4}(.8)^4(.2)+(.8)^5=.73728$. The mean working count is four, but this does not imply a probability of one-half for exceeding a threshold of four.

4.2 Geometric and negative binomial: waiting times

For $0<p\le 1$, the trial number G of the first success has

$$\mathbb{P}(G=k)=(1-p)^{k-1}p,\quad \mathbb{P}(G>k)=(1-p)^k,\quad k=1,2,\dots.$$

Let $q=1-p$. Differentiating $\sum_{k\ge 0}q^k=(1-q)^{-1}$ gives $\sum_{k\ge 1}kq^{k-1}=(1-q)^{-2}$ and $\sum_{k\ge 2}k(k-1)q^{k-2}=2(1-q)^{-3}$. Consequently $\mathbb{E}G=1/p$, $\mathbb{E}[G(G-1)]=2q/p^2$, and $\operatorname{Var} G=q/p^2$.

Memorylessness is a ratio of tails: $\mathbb{P}(G>s+t\mid G>s)=q^{s+t}/q^s=q^t$ for integer $s,t\ge 0$. A failure-count convention uses $G-1$; its mean is q/p and support begins at zero.

The trial number T_r of the rth success has

$$\mathbb{P}(T_r=k)=\binom{k-1}{r-1}p^r q^{k-r},\quad k\ge r.$$

The final trial must be a success; choose the other $r-1$ successes among the first $k-1$ trials. Alternatively T_r is the sum of r independent geometric waiting times, giving mean r/p and variance rq/p^2.

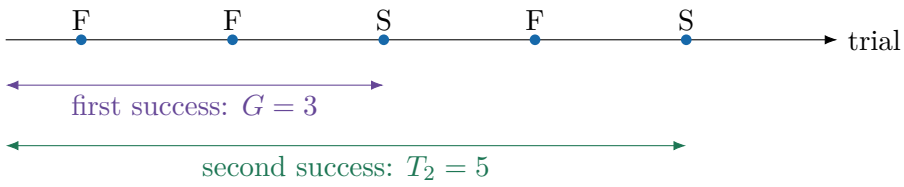


Figure 4.2: The observed sequence is F, F, S, F, S. Waiting times count trials, not the horizontal plotting coordinate.

Worked Example: Waiting time versus success count

With $p = .25$, the probability the third success occurs at trial six is $\binom{5}{2}(.25)^3(.75)^3$. The probability of three successes in six trials is instead $\binom{6}{3}(.25)^3(.75)^3$. The last trial is constrained only in the first question. For the waiting time, mean 12 and variance 36 follow from sums of geometric variables.

4.3 Hypergeometric: finite sampling without replacement

From N items of which K are successes, sample n uniformly without replacement. Counting subsets gives

$$\mathbb{P}(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \quad \max(0, n - N + K) \le k \le \min(n, K).$$

Set $p = K/N$ and let I_i indicate success on draw i. Then $\mathbb{E}I_i = p$, but for $i \neq j$,

$$\operatorname{Cov}(I_i, I_j) = \frac{K(K-1)}{N(N-1)} - p^2 = -\frac{p(1-p)}{N-1}.$$

Hence, for $N > 1$,

$$\mathbb{E}X = np, \quad \operatorname{Var} X = np(1-p)\frac{N-n}{N-1}.$$

The finite-population factor reduces uncertainty: drawing the entire population makes $X = K$ deterministic.

Worked Example: Sampling a finite batch

In a batch of twelve items, four are marked. Sampling three gives $\mathbb{P}(X = 2) = \binom{4}{2}\binom{8}{1}/\binom{12}{3} = 12/55$. Mean is one; variance is $3(1/3)(2/3)(9/11) = 6/11$. A binomial variance $2/3$ incorrectly ignores negative dependence.

4.4 Poisson and multinomial models

If $X \sim \text{Pois}(\lambda)$, $\mathbb{P}(X = k) = \mathrm{e}^{-\lambda}\lambda^k/k!$ for $k \ge 0$. The exponential series normalizes the law. Shifting the sum gives $\mathbb{E}X = \lambda$ and $\mathbb{E}[X(X-1)] = \lambda^2$, hence variance λ. The binomial limit follows for fixed k when $n \to \infty$, $p = \lambda/n$:

$$\binom{n}{k}(\lambda/n)^k(1-\lambda/n)^{n-k} \longrightarrow \mathrm{e}^{-\lambda}\lambda^k/k!.$$

A Poisson count assumption is not justified solely because the mean is known. Approximation requires an appropriate rare-event model.

For n independent categorical trials with probabilities $p_1, \ldots, p_d$, the count vector is multinomial:

$$\mathbb{P}(N_1 = n_1, \ldots, N_d = n_d) = \frac{n!}{\prod_i n_i!}\prod_i p_i^{n_i}, \quad \sum_i n_i = n.$$

Each count is binomial, with mean np_i and variance $np_i(1-p_i)$. Since one trial cannot belong to two categories, $\operatorname{Cov}(N_i, N_j) = -np_ip_j$ for $i \neq j$.

Worked Example: Independent counts conditioned on their sum

For independent $X \sim \text{Pois}(a)$ and $Y \sim \text{Pois}(b)$, convolution gives $X + Y \sim \text{Pois}(a + b)$. Dividing the joint probability by this sum's mass shows

$$\mathbb{P}(X = k \mid X + Y = n) = \binom{n}{k} \left(\frac{a}{a+b}\right)^k \left(\frac{b}{a+b}\right)^{n-k}.$$

Unconditional independence does not survive conditioning on the fixed total.

Standard Workflow: Select a discrete model

Fixed independent equal-probability trials: binomial. Wait for a success or several: geometric or negative binomial. Sample without replacement: hypergeometric. Rare independent occurrences: consider Poisson. Several categories per trial: multinomial. State the support and parameter convention before using a table.

4.5 Chapter Mind Map

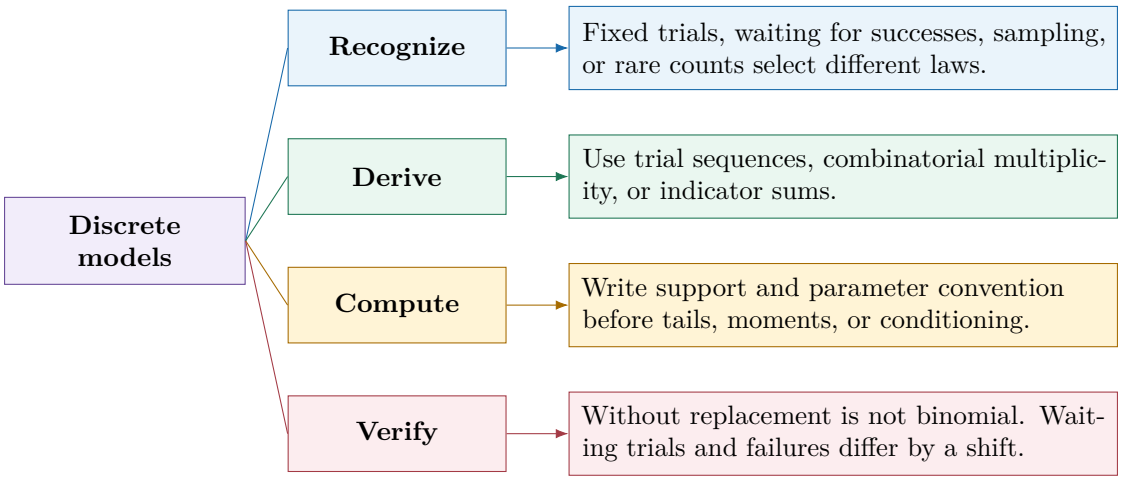


Figure 4.3: The chapter's model, calculation route, interpretation, and checks.

5 Principal Continuous Distributions

Source map: Ross 5.1–5.6; P5.1–P5.6.

5.1 Uniform models and geometric probability

For $a < b$, $X \sim \text{Unif}(a, b)$ has $f(x) = 1/(b - a)$ on (a, b), mean $(a + b)/2$, and variance $(b - a)^2/12$. These follow from integrating x and x^2 over the interval. Uniformity is relative to the stated coordinate or measure: uniform radius is not uniform area in a disk.

Worked Example: Uniform in a disk is not uniform in radius

A point is chosen uniformly by area in a disk of radius R. For $0 \le r \le R$, $\mathbb{P}(\text{distance} \le r) = \pi r^2/(\pi R^2) = r^2/R^2$, so the radius has density $2r/R^2$, mean $2R/3$, and second moment $R^2/2$. A uniform radius would have mean $R/2$ and overrepresent the center. The support and sampling measure determine the law.

5.2 Exponential survival and its memoryless property

For $\lambda > 0$, the rate-parameter exponential law has $f(x) = \lambda e^{-\lambda x}$ for $x > 0$, survival $S(x) = e^{-\lambda x}$, and CDF $1 - S(x)$. Integration by parts gives $\mathbb{E}X = 1/\lambda$ and $\mathbb{E}X^2 = 2/\lambda^2$, hence variance $1/\lambda^2$. Its hazard rate $h(x) = f(x)/S(x)$ is the constant λ.

The tail ratio proves memorylessness: $\mathbb{P}(X > s + t \mid X > s) = e^{-\lambda t}$. Conversely, a continuous nondegenerate survival function satisfying $S(s + t) = S(s)S(t)$ has exponential form: $-\log S$ is an additive monotone function, hence linear. It is not correct to use memorylessness for an arbitrary lifetime law.

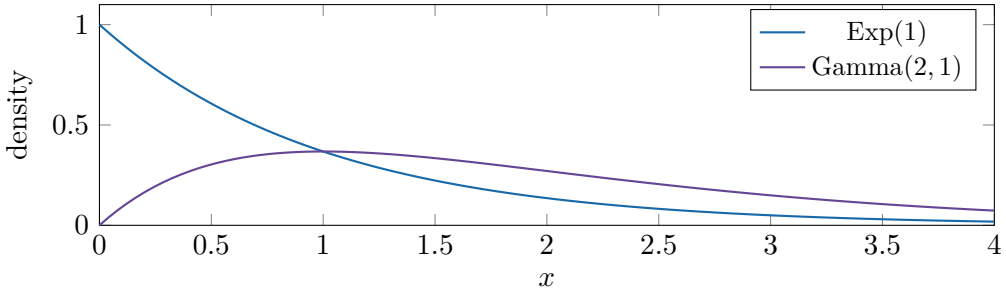


Figure 5.1: An exponential wait and a two-stage gamma wait have different shapes, despite both taking positive values.

Worked Example: Competing independent clocks

Let $X \sim \text{Exp}(2)$ and $Y \sim \text{Exp}(3)$ independently. Then $\mathbb{P}(\min(X, Y) > t) = e^{-5t}$, so the first arrival has rate five. Also

$$\mathbb{P}(X < Y) = \int_0^\infty 2e^{-2x}\mathbb{P}(Y > x)dx = \int_0^\infty 2e^{-5x}dx = 2/5.$$

The smaller-rate clock is less likely to win. The ratio is a consequence of independence, not a general rule for arbitrary waiting times.

5.3 Normal distribution: normalization and standardization

The standard normal density is $\phi(z) = (2\pi)^{-1/2}e^{-z^2/2}$. To derive its constant, square $I = \int_{\mathbb{R}} e^{-x^2/2}dx$. Polar coordinates give $I^2 = \int_0^{2\pi}\int_0^\infty e^{-r^2/2}r\,dr\,d\theta = 2\pi$. Symmetry gives mean zero; integration by parts using $\phi'(z) = -z\phi(z)$ yields second moment one.

For $X = \mu + \sigma Z$, $\sigma > 0$, the density is $\phi((x - \mu)/\sigma)/\sigma$, mean μ, and variance σ^2. Therefore

$$\mathbb{P}(a < X \le b) = \Phi((b - \mu)/\sigma) - \Phi((a - \mu)/\sigma).$$

The normal CDF has no elementary antiderivative. An exact answer in terms of Φ is valid; a decimal requires a table or numerical evaluation.

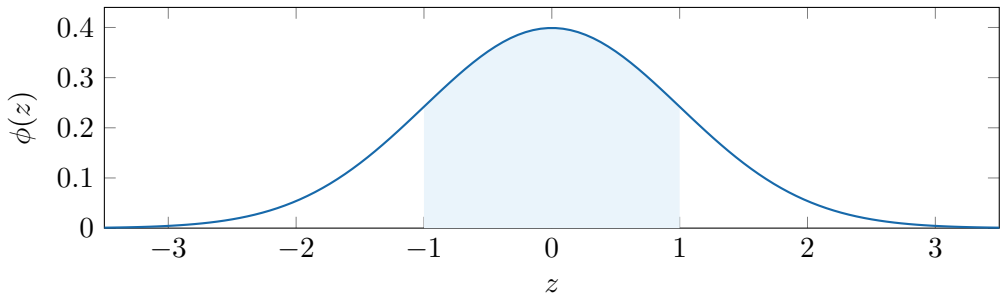


Figure 5.2: The shaded area is $\mathbb{P}(-1 \le Z \le 1) = 2\Phi(1) - 1$, not the height of the curve.

Worked Example: Normal threshold and inverse threshold

For $X \sim N(50, 9)$, the standard deviation is three. Then $\mathbb{P}(47 < X < 56) = \Phi(2) - \Phi(-1) \approx .8186$. The threshold exceeded by 5% of observations is $50 + 3\Phi^{-1}(.95) \approx 54.935$. Dividing by variance nine instead of standard deviation three changes the event and is incorrect.

5.4 Gamma, beta, and related families

Define $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1}\mathrm{e}^{-u}\mathrm{d}u$ for $\alpha > 0$. Integration by parts gives $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$ and $\Gamma(n) = (n-1)!$. A gamma law with shape α and rate λ has

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1}\mathrm{e}^{-\lambda x}, \quad x > 0.$$

Substitution $u = \lambda x$ normalizes it and gives

$$\mathbb{E}[X^r] = \frac{\Gamma(\alpha + r)}{\Gamma(\alpha)\lambda^r} \quad (r > -\alpha), \qquad \mathbb{E}X = \alpha/\lambda, \quad \operatorname{Var} X = \alpha/\lambda^2.$$

An integer-shape gamma is a sum of independent equal-rate exponential waits. A chi-square with ν degrees of freedom is Gamma$(\nu/2, 1/2)$; Chapter 10 connects it to squared normal variables.

The beta density on $(0, 1)$ is $x^{\alpha-1}(1-x)^{\beta-1}/B(\alpha, \beta)$. The identity $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$ follows by changing independent gamma integration variables to their sum and proportion (Chapter 7). Ratios of beta integrals yield

$$\mathbb{E}X = \frac{\alpha}{\alpha+\beta}, \quad \operatorname{Var} X = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Worked Example: A probability-valued random parameter

Let $P \sim \text{Beta}(2, 3)$. Since $B(2, 3) = 1/12$, $f_P(p) = 12p(1-p)^2$. Mean is $2/5$, variance $1/25$, and

$$\mathbb{P}(P > 1/2) = \left[6p^2 - 8p^3 + 3p^4\right]_{1/2}^{1} = 5/16.$$

The distribution describes variation in a probability parameter, not a count of successes.

5.5 Heavy tails and transformed positive laws

For a Pareto law with threshold $x_m > 0$ and shape $\alpha > 0$, $f(x) = \alpha x_m^\alpha / x^{\alpha+1}$ for $x \geq x_m$. Direct integration gives $\mathbb{E}[X^r] = \alpha x_m^r/(\alpha - r)$ only when $r < \alpha$; the moment diverges for $r \geq \alpha$. Thus a finite mean need not imply finite variance.

If $Y = \mathrm{e}^Z$ with $Z \sim N(\mu, \sigma^2)$, then Y is lognormal, with density $\phi((\log y - \mu)/\sigma)/(\sigma y)$ for $y > 0$. Completing the normal square yields $\mathbb{E}[Y^r] = \mathrm{e}^{r\mu + r^2\sigma^2/2}$. Despite all positive moments existing, its MGF is infinite for every $t > 0$; moment existence alone does not ensure an MGF near zero.

Standard Workflow: Continuous-law calculations

State support and parameters; integrate the density for probabilities; use a survival function for tails; standardize normals; recognize gamma/beta integrals; test convergence before quoting moments. Keep rate λ distinct from scale $1/\lambda$.

5.6 Chapter Mind Map

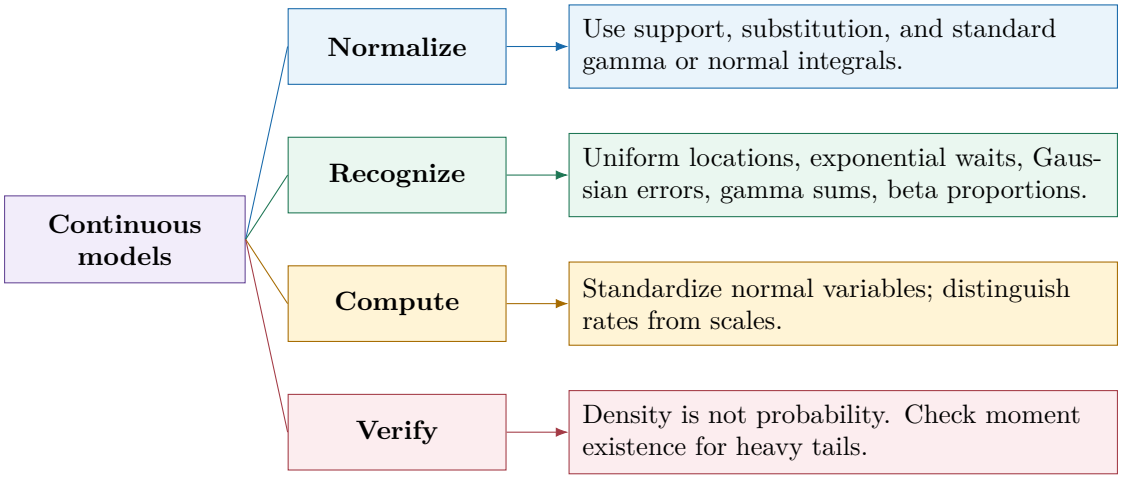


Figure 5.3: The chapter's model, calculation route, interpretation, and checks.

6 Expectation, Moments, and Inequalities

Source map: Ross 4.3–4.5, 7.2–7.4, 7.9, 8.2 and 8.5; P6.1–P6.6.

6.1 Expectation is a weighted integral, not an outcome prediction

For any law, write $\mathbb{E}[g(X)] = \int g \mathrm{d}F_X$. This is the law of the unconscious statistician (LOTUS): evaluate the function on the original variable, rather than deriving a new distribution unnecessarily. In a discrete model it is a weighted sum; in a density model it is an integral; in a mixed model it is both.

For a real variable, define positive and negative parts $X^+ = \max(X, 0)$ and $X^- = \max(-X, 0)$. A finite expectation requires $\mathbb{E}|X| < \infty$. If both $\mathbb{E}X^+$ and $\mathbb{E}X^-$ are infinite, the mean is undefined, not zero by symmetry. Finite second moment implies finite first absolute moment by Cauchy–Schwarz.

Key Result: Moments, variance, and affine changes

The rth raw moment is $\mathbb{E}X^r$; the central moment is $\mathbb{E}[(X-\mu)^r]$. For finite second moment,

$$\operatorname{Var} X = \mathbb{E}[(X-\mu)^2] = \mathbb{E}X^2 - \mu^2, \quad \mathbb{E}(aX+b) = a\mu + b, \quad \operatorname{Var}(aX+b) = a^2 \operatorname{Var} X.$$

Skewness is $\mathbb{E}[(X-\mu)^3]/\sigma^3$, and kurtosis is $\mathbb{E}[(X-\mu)^4]/\sigma^4$ when defined. These dimensionless summaries do not uniquely determine a general law.

The variance identity follows by expanding $X^2 - 2\mu X + \mu^2$ and taking expectations. Linearity follows from linearity of sums or integrals; it does *not* require independence. By contrast $\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y$ requires a justification, such as independence with integrability.

Worked Example: Transform without finding a new density

For $U \sim \mathrm{Unif}(0,2)$, $\mathbb{E}[(U-1)^2] = \frac{1}{2}\int_0^2 (u-1)^2 \mathrm{d}u = 1/3$. Since $\mathbb{E}U = 1$, this is also $\operatorname{Var} U$. Computing the law of $(U-1)^2$ is possible but adds unnecessary inverse branches if only this mean is requested.

6.2 Indicators and tail sums

An indicator satisfies $\mathbb{E}\mathbf{1}_{\{A\}} = \mathbb{P}(A)$. If $N = \sum_i \mathbf{1}_{\{A_i\}}$, then $\mathbb{E}N = \sum_i \mathbb{P}(A_i)$ whether or not the events are independent. For nonnegative integer N, the pointwise identity $N = \sum_{k\geq 0} \mathbf{1}_{\{N>k\}}$ and monotone interchange give $\mathbb{E}N = \sum_{k\geq 0} \mathbb{P}(N > k)$. For nonnegative real X, $X = \int_0^\infty \mathbf{1}_{\{X>t\}} \mathrm{d}t$ yields

$$\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t)\mathrm{d}t, \quad \mathbb{E}X^r = r\int_0^\infty t^{r-1}\mathbb{P}(X > t)\mathrm{d}t \quad (r > 0).$$

The nonnegative integrands justify interchange even when the answer is infinite.

Worked Example: Expected occupied boxes

Place m labeled balls independently into n boxes with equal probabilities. Let I_j indicate that box j is nonempty. It is empty with probability $(1 - 1/n)^m$, so

$$\mathbb{E}\left[\sum_j I_j\right] = n[1 - (1 - 1/n)^m].$$

The indicators are dependent; this does not obstruct the expectation. It would matter if computing the variance.

6.3 Markov, Chebyshev, and one-sided bounds

For $X \geq 0$ and $a > 0$, $X \geq a\mathbf{1}_{\{X\geq a\}}$. Taking expectations proves $\mathbb{P}(X \geq a) \leq \mathbb{E}X/a$. Apply Markov to $(X - \mu)^2$ to obtain Chebyshev:

$$\mathbb{P}(|X - \mu| \geq a) \leq \min(1, \sigma^2/a^2).$$

For $W = X - \mu$ and $b \geq 0$, $W \geq a$ implies $(W + b)^2 \geq (a + b)^2$. Markov gives a bound $(\sigma^2 + b^2)/(a + b)^2$. Differentiating in b yields the minimizer $b = \sigma^2/a$, proving Cantelli's one-sided inequality

$$\mathbb{P}(X - \mu \geq a) \leq \frac{\sigma^2}{\sigma^2 + a^2}.$$

Worked Example: Different information gives different bounds

A nonnegative variable has mean ten and variance four. Markov gives $\mathbb{P}(X \geq 14) \leq 10/14 = 5/7$. Chebyshev gives $\mathbb{P}(X \geq 14) \leq 4/16 = 1/4$. Cantelli improves the one-sided bound to $4/(4+16) = 1/5$. None of these bounds claims the actual probability is equal to the number shown.

6.4 Jensen and Cauchy–Schwarz with their assumptions

For convex g on an interval containing the support, a supporting line at an interior mean μ gives $g(x) \geq g(\mu) + s(x - \mu)$. Taking expectations proves $g(\mathbb{E}X) \leq \mathbb{E}g(X)$ when expectations are meaningful (finite in the elementary statement). For differentiable g, $s = g'(\mu)$; for strictly convex g, equality forces X to be constant almost surely. Concave functions reverse the inequality.

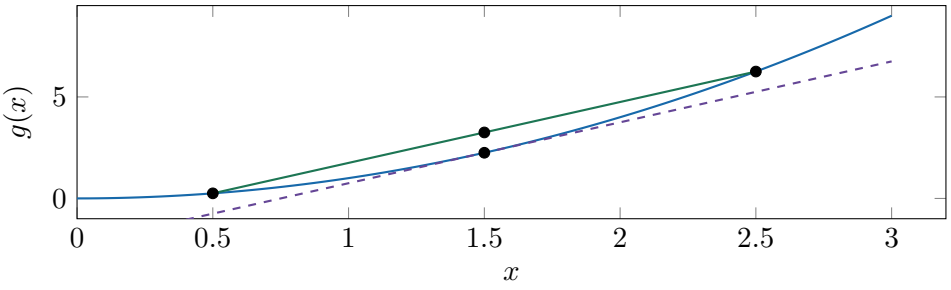


Figure 6.1: For $g(x) = x^2$, the tangent lies below the graph. The chord average lies above the value at the average.

Worked Example: A nonlinear mean cannot be substituted blindly

For $X > 0$ with finite $\mathbb{E}X$ and $\mathbb{E}(1/X)$, convexity of $1/x$ gives $\mathbb{E}(1/X) \geq 1/\mathbb{E}X$. For $X \sim \text{Unif}(1,3)$ the two sides are $(\log 3)/2$ and $1/2$. Concavity of log similarly yields $\mathbb{E}\log X \leq \log \mathbb{E}X$. The direction changes because the curvature changes.

For square-integrable X, Y, $0 \leq \mathbb{E}[(X - tY)^2]$ is a quadratic in t. If $\mathbb{E}Y^2 > 0$, minimize at $t = \mathbb{E}(XY)/\mathbb{E}Y^2$ to obtain $|\mathbb{E}(XY)|^2 \leq \mathbb{E}X^2\mathbb{E}Y^2$; the zero case is immediate. Equality means $X = tY$ almost surely. Centering the variables gives the correlation bound in Chapter 9.

Standard Workflow: Moment or bound questions

First check existence. Then select direct LOTUS, indicators, a tail integral, or conditioning. For a bound, name the transformed nonnegative variable and threshold. For Jensen, write the function and verify its curvature on the entire support before taking expectations.

6.5 Chapter Mind Map

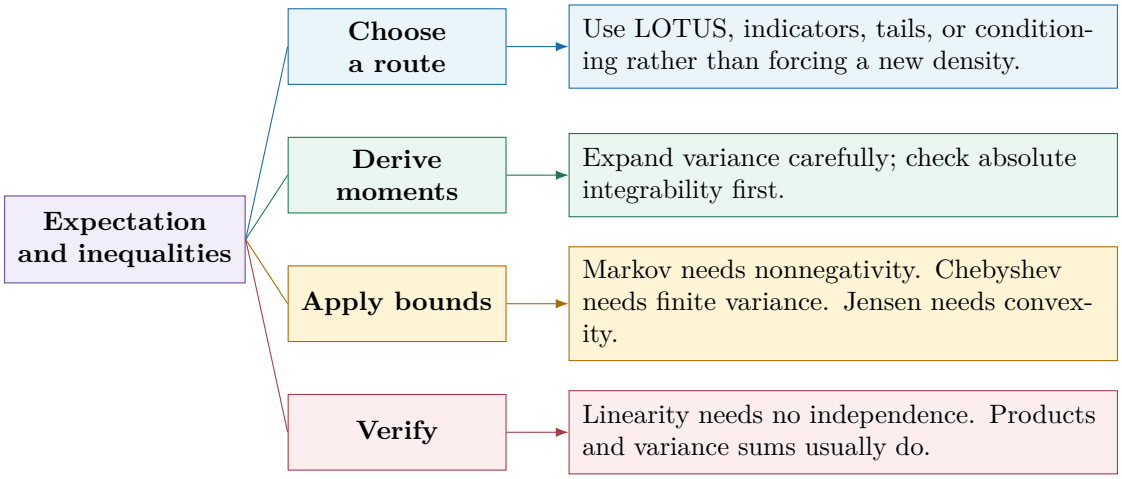


Figure 6.2: The chapter's model, calculation route, interpretation, and checks.

7 Transformations, Convolution, and Order Statistics

Source map: Ross 7.7, 6.3, 6.6–6.7; P7.1–P7.6.

7.1 The CDF method works beyond monotonicity

For $Y = g(X)$, begin with $F_Y(y) = \mathbb{P}(g(X) \leq y)$ and solve that inequality on the *actual support*. For increasing differentiable one-to-one g with inverse h, differentiation gives $f_Y(y) = f_X(h(y))|h'(y)|$. The absolute value also handles a decreasing transformation. When several inverse branches exist, the disjoint preimage intervals contribute additively:

$$f_Y(y) = \sum_{x:g(x)=y} \frac{f_X(x)}{|g'(x)|}$$

at regular values with isolated branches. Singular points can be handled by the CDF and occur on null sets in standard examples. Atoms must be transformed separately and merged if several atoms map to the same value.

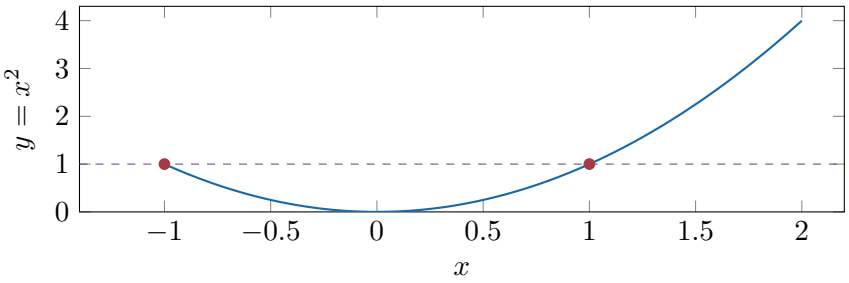


Figure 7.1: Squaring has two inverse branches. Omitting the negative root loses probability whenever the input can be negative.

Worked Example: A nonmonotone transform with unequal branch lengths

Let $X \sim \text{Unif}(-1, 2)$ and $Y = X^2$. For $0 < y < 1$, the preimage is $[-\sqrt{y}, \sqrt{y}]$, so $F_Y(y) = 2\sqrt{y}/3$. For $1 \le y < 4$, it is $[-1, \sqrt{y}]$, so $F_Y(y) = (1 + \sqrt{y})/3$. Thus $f_Y(y) = 1/(3\sqrt{y})$ on $(0, 1)$ and $1/(6\sqrt{y})$ on $(1, 4)$. Both pieces integrate to total mass one. The branch disappears at $y = 1$, changing the formula.

7.2 Independent sums and convolution

If independent continuous X, Y have densities, then conditioning on $X = x$ gives

$$F_{X+Y}(s) = \int f_X(x) F_Y(s - x) \mathrm{d}x, \quad f_{X+Y}(s) = \int f_X(x) f_Y(s - x) \mathrm{d}x.$$

The valid limits are the overlap of x in X's support and $s - x$ in Y's support. For independent discrete variables, replace the integral by $\sum_k p_X(k) p_Y(s - k)$. Without independence, use the actual joint density $\int f_{X,Y}(x, s - x) \mathrm{d}x$, not a product of marginals.

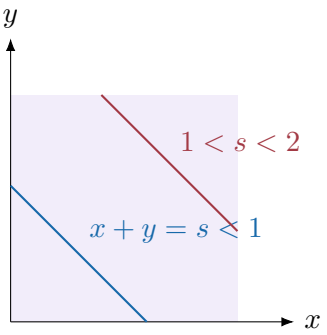


Figure 7.2: The line $x + y = s$ intersects different parts of the unit square when $s < 1$ and $s > 1$, producing a piecewise convolution.

Worked Example: The triangular sum of two uniforms

For independent $X, Y \sim \text{Unif}(0, 1)$, the convolution integrand is one when $\max(0, s - 1) < x < \min(1, s)$. Its interval length is s for $0 < s < 1$ and $2 - s$ for $1 \le s < 2$, zero otherwise. Symmetry gives mean one; variances add to $1/6$. In particular $\mathbb{P}(X + Y < 1/2) = \int_0^{1/2} s \mathrm{d}s = 1/8$.

7.3 Joint transformations: inverse map and Jacobian

For a differentiable one-to-one map $\boldsymbol{y} = g(\boldsymbol{x})$ on each relevant region, small volumes satisfy $\mathrm{d}\boldsymbol{x} = |\det Dg^{-1}(\boldsymbol{y})| \mathrm{d}\boldsymbol{y}$. Probability conservation gives

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = f_{\boldsymbol{X}}(g^{-1}(\boldsymbol{y})) |\det Dg^{-1}(\boldsymbol{y})|$$

on the transformed support. If the map is not globally one-to-one, sum the branch contributions. The Jacobian is a volume factor, not an optional normalization fix.

Worked Example: Gamma sum and beta proportion, fully factored

Let independent $X \sim \text{Gamma}(a, \lambda)$ and $Y \sim \text{Gamma}(b, \lambda)$. Define $S = X + Y$ and $R = X/(X+Y)$. Inverse: $x = sr$, $y = s(1-r)$, support $s > 0$, $0 < r < 1$. The absolute inverse determinant is s. Thus

$$\begin{aligned} f_{S,R}(s,r) &= \frac{\lambda^{a+b}}{\Gamma(a)\Gamma(b)} s^{a+b-1} \mathrm{e}^{-\lambda s} r^{a-1}(1-r)^{b-1} \\ &= \frac{\lambda^{a+b} s^{a+b-1} \mathrm{e}^{-\lambda s}}{\Gamma(a+b)} \frac{\Gamma(a+b) r^{a-1}(1-r)^{b-1}}{\Gamma(a)\Gamma(b)}. \end{aligned}$$

The support is a product, and each factor normalizes. Hence $S \sim \text{Gamma}(a+b, \lambda)$, $R \sim \text{Beta}(a, b)$, and S, R are independent. The normalization also proves the beta–gamma identity. Unequal rates destroy this factorization in general.

7.4 Minima, maxima, and ranks

For independent identically distributed continuous $X_1, \ldots, X_n$ with CDF F and density f, independence gives $F_{\max}(x) = F(x)^n$ and $\mathbb{P}(\min > x) = (1 - F(x))^n$. For the kth smallest value, exactly $k-1$ observations lie below a small interval near x, one lies inside, and $n-k$ above. Counting assignments and dividing by interval width gives

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} F(x)^{k-1}(1-F(x))^{n-k} f(x).$$

This density assumes no ties, as ensured by continuity. For discrete data, use a binomial count expression for the CDF rather than inserting a mass function into this derivative formula.

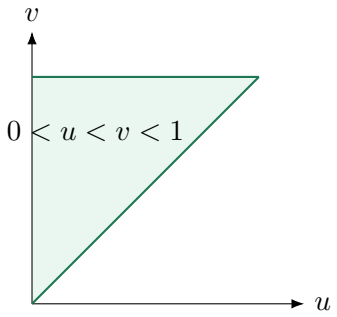


Figure 7.3: For two uniform observations, the joint support of minimum u and maximum v is the upper triangle; its density is two.

Worked Example: A uniform median and the sample range

For five independent uniforms, the third order statistic has density $30x^2(1-x)^2$ on $(0,1)$, a Beta$(3,3)$ law. Its mean is $1/2$ and variance $1/28$. For two independent uniforms, minimum and maximum have joint density two on $0 < u < v < 1$. Their range $R = v - u$ has density $\int_0^{1-r} 2\mathrm{d}u = 2(1-r)$ on $(0,1)$ and mean $1/3$.

Standard Workflow: Transformation questions

Write the map; find its image support; determine every inverse branch; choose CDF, convolution, or Jacobian; compute the density with the absolute volume factor; add any atoms; normalize and check a moment using the original variables when possible.

7.5 Chapter Mind Map

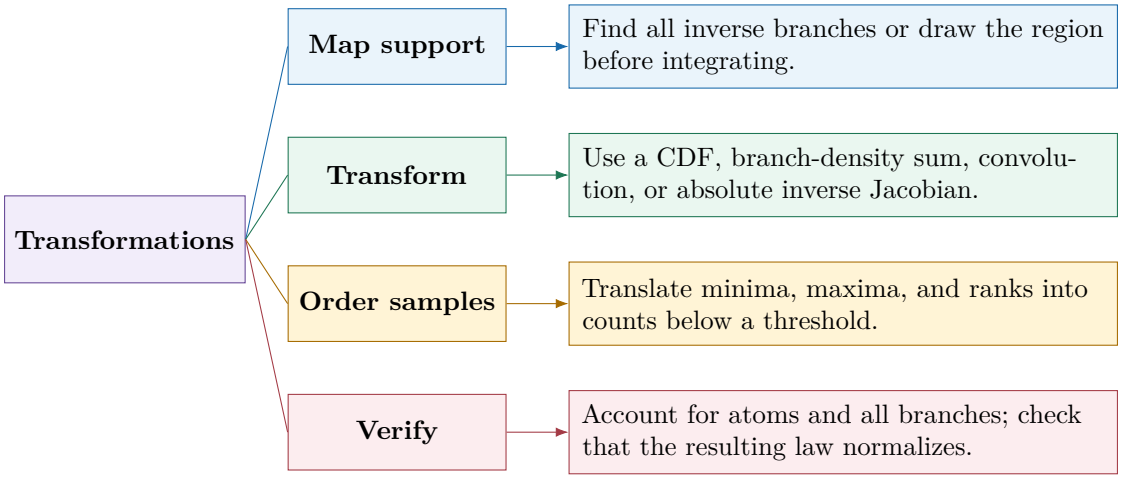


Figure 7.4: The chapter's model, calculation route, interpretation, and checks.

8 Joint Laws and Conditional Expectation

Source map: Ross 6.1–6.5, 7.5–7.6; P8.1–P8.6.

8.1 Joint CDFs and marginals

For (X, Y), $F(x, y) = \mathbb{P}(X \le x, Y \le y)$. Rectangle probabilities are the four-corner difference $F(b, d) - F(a, d) - F(b, c) + F(a, c)$ for $(a, b] \times (c, d]$. A valid joint CDF needs nonnegative rectangle increments; separate monotonicity alone is not enough. A joint density integrates to one over its support. Marginals are obtained by integrating over the other coordinate, with limits determined by the support.

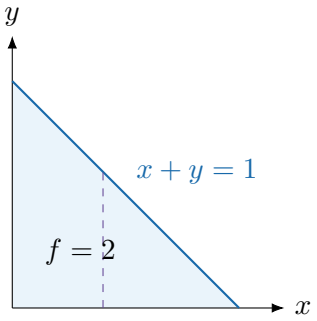


Figure 8.1: A uniform density on a triangle is not a product of uniform marginals. Vertical slices have length $1 - x$.

Worked Example: Normalize, marginalize, and test independence

Let $f_{X,Y}(x, y) = c$ on $x > 0, y > 0, x + y < 1$. The triangle area is $1/2$, so $c = 2$. Then $f_X(x) = 2(1 - x)$ and $f_Y(y) = 2(1 - y)$ on $(0, 1)$. The variables are dependent: the marginal product is positive at $(.8, .8)$, but the joint density is zero there. An algebraic factorization inside the triangle cannot repair its nonproduct support.

Independence is equivalent to factorization of the joint CDF at every point; for discrete or absolutely continuous laws it is equivalent to joint mass/density factorization (almost everywhere for densities). Functions of independent variables remain independent when each function uses only its own variable, because inverse-image events still factor.

8.2 Conditional distributions are normalized slices

For discrete laws, $p_{X|Y}(x \mid y) = p_{X,Y}(x,y)/p_Y(y)$ when $p_Y(y) > 0$. For a density, define $f_{X|Y}(x \mid y) = f_{X,Y}(x,y)/f_Y(y)$ for almost every y with $f_Y(y) > 0$. The event $Y = y$ has probability zero in the continuous case, so this is not a quotient of two zero event probabilities. It is the density of a conditional law, justified by the joint factorization and disintegration in this setting.

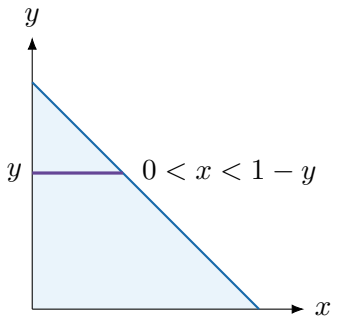


Figure 8.2: For $X > 0, Y > 0, X + Y < 1$, conditioning on $Y = y$ gives $0 < x < 1 - y$. The conditional density is $1/(1-y)$.

Worked Example: Conditional moments on a triangle

Continue the uniform triangle $x + y < 1$. For $0 < y < 1$,

$$f_{X|Y}(x \mid y) = 1/(1-y), \quad 0 < x < 1 - y.$$

Thus $X \mid Y = y$ is uniform on $(0, 1-y)$, with mean $(1-y)/2$ and variance $(1-y)^2/12$. In particular the conditional probability $\mathbb{P}(X < .2 \mid Y = .7)$ is $2/3$, even though the conditioning event has probability zero under the marginal density.

8.3 The tower rule and total variance

Define $m(Y) = \mathbb{E}[X \mid Y]$ by integrating x against each conditional law. In the density case,

$$\mathbb{E}[m(Y)] = \int \left(\int x \frac{f_{X,Y}(x,y)}{f_Y(y)} \mathrm{d}x \right) f_Y(y) \mathrm{d}y = \mathbb{E}X$$

when X is integrable; the discrete proof replaces integrals by sums. More generally $\mathbb{E}[h(Y)X] = \mathbb{E}[h(Y)m(Y)]$ whenever the products are integrable.

For square-integrable X, decompose $X - \mu = (X - m(Y)) + (m(Y) - \mu)$. The cross term has mean zero because its conditional expectation given Y is zero. Squaring gives

Key Result: Total variance

$$\operatorname{Var} X = \mathbb{E}[\operatorname{Var}(X \mid Y)] + \operatorname{Var}(\mathbb{E}[X \mid Y]).$$

Within-group variability and between-group variability are different contributions. Averaging conditional variances alone loses the second term.

Worked Example: A random success probability

Let $P \sim \mathrm{Unif}(0,1)$ and, conditional on $P = p$, let $N \sim \mathrm{Bin}(n,p)$. Then

$$\mathbb{E}N = n/2, \quad \operatorname{Var} N = n\mathbb{E}[P(1-P)] + n^2 \operatorname{Var} P = n/6 + n^2/12 = n(n+2)/12.$$

Integrating the binomial mass gives $\mathbb{P}(N = k) = \binom{n}{k} B(k+1, n-k+1) = 1/(n+1)$ for $k = 0, \ldots, n$. A mixture of binomials is not usually binomial with the averaged probability; latent heterogeneity increases variance.

8.4 Prediction as an orthogonal decomposition

For any square-integrable predictor $g(Y)$, write $X - g(Y) = (X - m(Y)) + (m(Y) - g(Y))$. Conditioning kills the cross term, so

$$\mathbb{E}[(X - g(Y))^2] = \mathbb{E}[\mathrm{Var}(X \mid Y)] + \mathbb{E}[(m(Y) - g(Y))^2].$$

The second term is nonnegative and vanishes exactly when $g(Y) = m(Y)$ almost surely. Thus conditional expectation is the minimum mean-square predictor. This is the probability analogue of orthogonal projection in Algebra 2M; the inner product is $\langle U, V \rangle = \mathbb{E}(UV)$.

Worked Example: A nonlinear optimal predictor

Let $Y \sim \mathrm{Unif}(-1, 1)$ and $X = Y^2 + \varepsilon$, where ε is independent, centered, and has variance s^2. Then $\mathbb{E}[X \mid Y] = Y^2$, with mean-square error s^2. Because $\mathrm{Cov}(Y, Y^2) = 0$, the best affine predictor from Y is the constant $1/3$, with larger error $s^2 + \mathrm{Var}(Y^2) = s^2 + 4/45$. Zero correlation does not mean that Y contains no predictive information.

Standard Workflow: Joint-law questions

Draw the support; normalize; compute both marginals; test independence on the full domain; divide by the correct marginal for conditioning; calculate conditional moments; use tower and total variance; check each conditional law integrates to one.

8.5 Chapter Mind Map

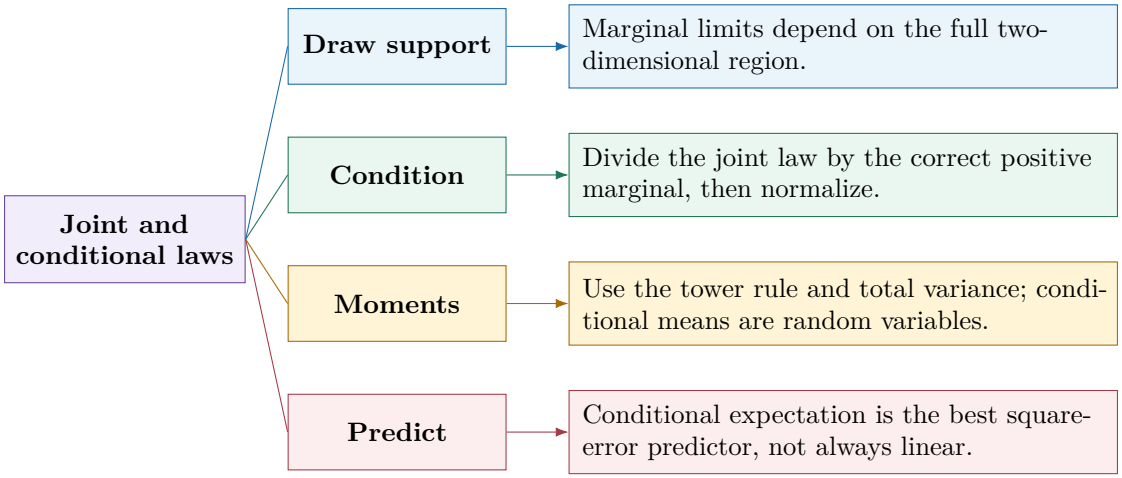


Figure 8.3: The chapter's model, calculation route, interpretation, and checks.

9 Covariance, Correlation, and Random Vectors

Source map: Ross 7.4–7.6, 7.8; P9.1–P9.6.

9.1 Cross terms carry dependence

For square-integrable variables, $\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}(XY) - \mu_X \mu_Y$. Bilinearity follows by expanding centered sums. In particular,

$$\mathrm{Var}\left(\sum_i a_i X_i\right) = \sum_i a_i^2 \,\mathrm{Var}\, X_i + 2 \sum_{i<j} a_i a_j \,\mathrm{Cov}(X_i, X_j).$$

Independence makes cross covariances zero, but pairwise uncorrelatedness is enough for this variance-sum formula. It is not enough for distributional factorization or for every application of a limit theorem.

Worked Example: Variance of occupied boxes

For m balls independently and uniformly assigned to $n \geq 2$ boxes, let N count nonempty boxes and put $q = (1-1/n)^m$, $r = (1-2/n)^m$. For different boxes, inclusion–exclusion gives $\mathbb{E}(I_i I_j) = 1-2q+r$, so $\operatorname{Cov}(I_i, I_j) = r - q^2$. Therefore

$$\operatorname{Var} N = n(1-q)q + n(n-1)(r-q^2).$$

The covariance correction is generally negative. An incorrect independent-indicator approximation drops it while leaving the expectation unchanged.

9.2 Correlation and its exact equality cases

For positive standard deviations, $\rho = \operatorname{Cov}(X,Y)/(\sigma_X \sigma_Y)$. Cauchy–Schwarz for centered variables gives $|\rho| \leq 1$. Equality holds if and only if $Y - \mu_Y = c(X - \mu_X)$ almost surely for a nonzero constant c; the sign of c is the sign of ρ. Correlation is undefined when either variance is zero. It does not measure every form of dependence.

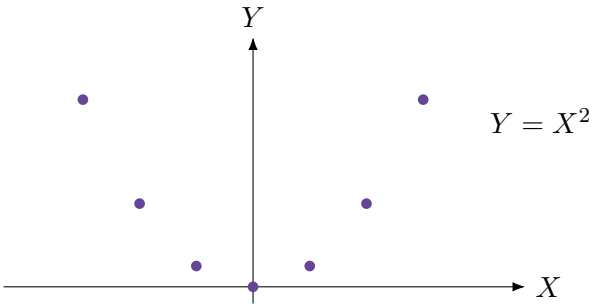


Figure 9.1: A symmetric X and $Y = X^2$ can be uncorrelated although Y is completely determined by X.

Worked Example: Uncorrelated but dependent

For $X \sim \text{Unif}(-1,1)$ and $Y = X^2$, $\mathbb{E}X = 0$, $\mathbb{E}X^3 = 0$, so $\operatorname{Cov}(X,Y) = 0$. But $\mathbb{P}(Y < 1/4 \mid |X| < 1/2) = 1$, while $\mathbb{P}(Y < 1/4) = 1/2$. The dependence is nonlinear. Both variables have positive variance, so this is a genuine zero-correlation example, not a degenerate exception.

9.3 Covariance matrices and linear maps

For a column vector $\boldsymbol{X}$ with mean $\boldsymbol{\mu}$, its covariance matrix is $\Sigma = \mathbb{E}[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^{\mathsf{T}}]$. It is symmetric. For every $\boldsymbol{a}$,

$$\boldsymbol{a}^{\mathsf{T}} \Sigma \boldsymbol{a} = \operatorname{Var}(\boldsymbol{a}^{\mathsf{T}} \boldsymbol{X}) \geq 0,$$

so it is positive semidefinite, not necessarily invertible. A zero quadratic form means that linear combination is constant almost surely. If $\boldsymbol{Y} = A\boldsymbol{X} + \boldsymbol{b}$, center both sides to derive

$$\mathbb{E}\boldsymbol{Y} = A\boldsymbol{\mu} + \boldsymbol{b}, \quad \operatorname{Cov}(\boldsymbol{Y}) = A\Sigma A^{\mathsf{T}}, \quad \operatorname{Cov}(\boldsymbol{X}, \boldsymbol{Y}) = \Sigma A^{\mathsf{T}}.$$

This is a basis-change calculation linked directly to linear algebra; an orthogonal diagonalization removes cross covariance but does not in general create independence.

Worked Example: A transformed pair

Let $\Sigma = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}$ and $U = X + Y$, $V = 2X - Y$. With $A = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$,

$$\operatorname{Var} U = 15, \quad \operatorname{Var} V = 21, \quad \operatorname{Cov}(U, V) = 8 + 1 - 9 = 0.$$

Thus $\operatorname{Cov}(U, V)$ is diagonal, but independence still needs a joint-law assumption such as joint Gaussianity.

9.4 Best linear prediction and random sums

For $\operatorname{Var} X > 0$, minimize $\mathbb{E}[(Y - a - bX)^2]$. Differentiating in a yields $a = \mu_Y - b\mu_X$; substituting gives $\sigma_Y^2 - 2b \operatorname{Cov}(X, Y) + b^2\sigma_X^2$. Its minimum occurs at $b = \operatorname{Cov}(X, Y)/\sigma_X^2$, with error $\sigma_Y^2(1 - \rho^2)$. This optimum among affine predictors need not equal the optimal nonlinear conditional expectation.

If N is a nonnegative integer independent of an iid sequence (X_i) with mean μ and variance σ^2, define $S = \sum_{i=1}^{N} X_i$, with an empty sum zero. Conditioning gives $\mathbb{E}[S \mid N] = N\mu$ and $\operatorname{Var}(S \mid N) = N\sigma^2$. Hence

$$\mathbb{E}S = \mu\mathbb{E}N, \quad \operatorname{Var} S = \sigma^2\mathbb{E}N + \mu^2 \operatorname{Var} N.$$

Worked Example: Compound claims

If $N \sim \text{Pois}(4)$ and independent claim sizes have mean ten and variance nine, the aggregate has mean forty and variance $4(9) + 4(100) = 436$. Keeping only the conditional variance contribution would give thirty-six and miss variability in the number of claims.

Standard Workflow: Covariance and vector calculations

Record the mean vector and covariance matrix. Form the transformation matrix from the requested linear combinations. Compute $A\Sigma A^{\mathsf{T}}$, retain all cross terms, and test nonnegative diagonal entries and determinants. Claim independence only after examining the joint distribution.

9.5 Chapter Mind Map

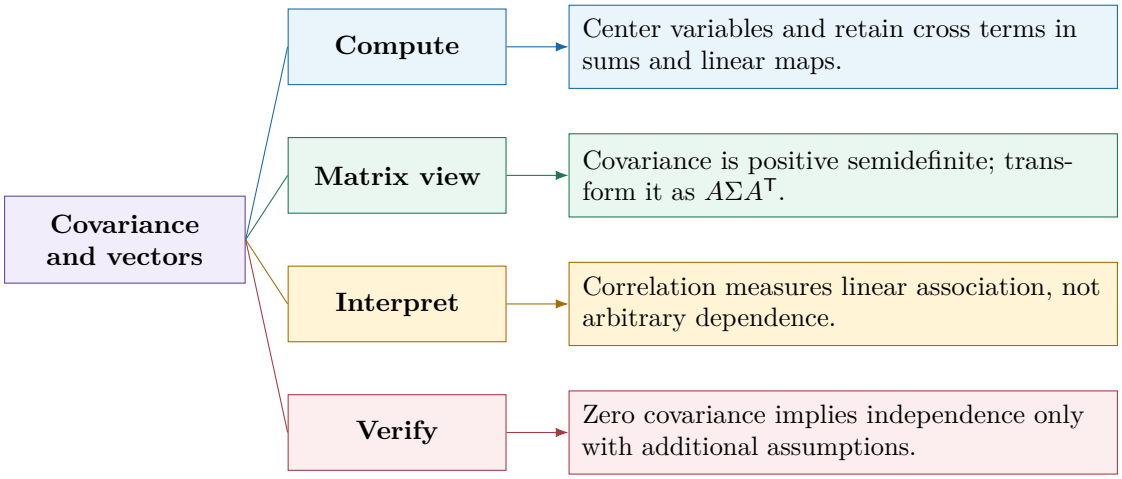


Figure 9.2: The chapter's model, calculation route, interpretation, and checks.

10 Gaussian Vectors and Conditional Normal Laws

Source map: Ross 5.4 and 7.8; vector derivations expanded to match the syllabus; P10.1–P10.6.

10.1 Construction, density, and singular cases

Let $\boldsymbol{Z}$ have independent $N(0,1)$ entries. An affine image $\boldsymbol{X} = \boldsymbol{\mu} + B\boldsymbol{Z}$ is Gaussian, with covariance $\Sigma = BB^{\mathsf{T}}$. Equivalently every linear combination is univariate normal (allowing constants). If Σ is positive definite in dimension d, an invertible square root exists; the Jacobian of $\boldsymbol{z} = \Sigma^{-1/2}(\boldsymbol{x} - \boldsymbol{\mu})$ gives

Key Result: Multivariate normal density

$$f(\boldsymbol{x}) = \frac{\exp[-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^{\mathsf{T}}\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu})]}{(2\pi)^{d/2}\sqrt{\det \Sigma}}.$$

If Σ is singular, this full-dimensional density formula is invalid. The Gaussian law still exists but is supported on a lower-dimensional affine subspace.

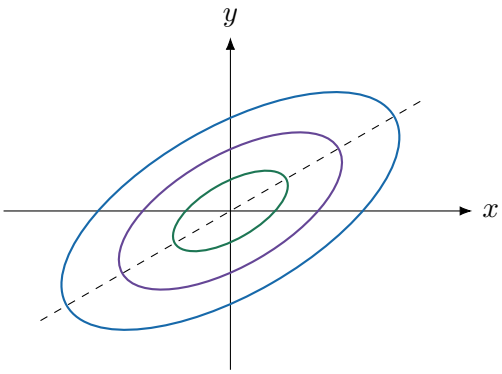


Figure 10.1: Gaussian density contours are ellipses aligned with covariance eigenvectors. Eigenvalues set the squared axis scales.

For positive variances and $|\rho| < 1$, the bivariate exponent is

$$\frac{1}{2(1-\rho^2)}\left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right].$$

Setting $\rho = 0$ factors the joint density into marginals. More generally, Gaussian blocks with zero cross covariance are independent, including singular blocks, as follows from the factorization of their joint characteristic function. Having normal marginals alone is insufficient to use this rule. Indeed the affine construction and independence of the standard normal coordinates give, for every real column vector $\boldsymbol{t}$,

$$\varphi_{\boldsymbol{X}}(\boldsymbol{t}) = \mathbb{E}e^{i\boldsymbol{t}^{\mathsf{T}}\boldsymbol{X}} = e^{i\boldsymbol{t}^{\mathsf{T}}\boldsymbol{\mu}}\prod_j e^{-[(B^{\mathsf{T}}\boldsymbol{t})_j]^2/2} = e^{i\boldsymbol{t}^{\mathsf{T}}\boldsymbol{\mu} - \boldsymbol{t}^{\mathsf{T}}\Sigma\boldsymbol{t}/2}.$$

If the cross covariance between two blocks is zero, this quadratic form has no mixed block term. The characteristic function therefore factors into the two marginal characteristic functions, proving independence by joint-transform uniqueness, without needing an invertible density.

Worked Example: Normal marginals but not a Gaussian pair

Let $Z \sim N(0,1)$ and an independent sign S be equally likely to be ± 1. Set $X = Z$, $Y = SZ$. Both marginals are standard normal and $\mathbb{E}(XY) = \mathbb{E}S\mathbb{E}Z^2 = 0$. But $|X| = |Y|$ almost surely, so they are dependent. The pair is not jointly Gaussian: $X + Y$ has an atom at zero, whereas a nonconstant normal variable has none.

10.2 Linear combinations and orthogonal normal coordinates

Every affine image remains Gaussian. Therefore a zero covariance computed after a linear transformation does establish independence *within this family.* If $\Sigma = Q\Lambda Q^{\mathsf{T}}$ with orthogonal Q and positive eigenvalues, then $\boldsymbol{W} = \Lambda^{-1/2}Q^{\mathsf{T}}(\boldsymbol{X} - \boldsymbol{\mu})$ has independent standard normal components. This is whitening: rotate to principal axes and scale by standard deviations, not by eigenvalues themselves.

Worked Example: Sum and difference of normal measurements

If X, Y are independent $N(0, \sigma^2)$, then $U = (X + Y)/\sqrt{2}$ and $V = (X - Y)/\sqrt{2}$ have variances σ^2 and covariance zero. Joint Gaussianity makes them independent. For unequal input variances, covariance is $(\operatorname{Var} X - \operatorname{Var} Y)/2$, so the unweighted sum and difference are not independent in general.

10.3 Conditional normal law: derive via an independent residual

For a jointly normal pair with $\sigma_X^2 > 0$, let $b = \operatorname{Cov}(X, Y)/\sigma_X^2$ and $R = Y - \mu_Y - b(X - \mu_X)$. Then $\operatorname{Cov}(R, X) = 0$. The pair (R, X) is Gaussian, hence independent, and $\operatorname{Var} R = \sigma_Y^2 - \operatorname{Cov}(X, Y)^2/\sigma_X^2$. It follows that

$$Y \mid X = x \sim N\left(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X),\ \sigma_Y^2(1 - \rho^2)\right).$$

At $|\rho| = 1$, residual variance is zero and the conditional law is a point mass, not an ordinary positive-variance density.

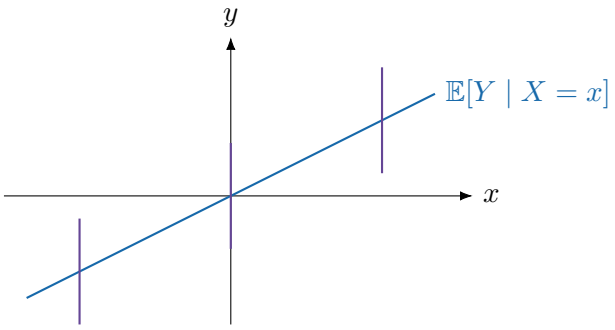


Figure 10.2: Conditional means lie on the regression line. Conditional spread is constant for a jointly Gaussian pair.

Worked Example: A conditional exceedance probability

Let means be $(2, -1)$ and covariance $\begin{pmatrix} 4 & 3 \\ 3 & 9 \end{pmatrix}$. Then $Y \mid X = 4$ is normal with mean $-1 + (3/4)(4 - 2) = 1/2$ and variance $9 - 9/4 = 27/4$. Thus

$$\mathbb{P}(Y > 2 \mid X = 4) = 1 - \Phi\left(\frac{2 - 1/2}{\sqrt{27}/2}\right) = 1 - \Phi(1/\sqrt{3}).$$

The conditional variance is smaller than the marginal variance nine, consistent with observing information.

For Gaussian blocks with covariance $\begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix}$ and invertible Σ_{XX}, the same residual argument gives conditional mean $\boldsymbol{\mu}_Y + \Sigma_{YX}\Sigma_{XX}^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_X)$ and covariance $\Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^{-1}\Sigma_{XY}$. This Schur complement must be positive semidefinite. It links block elimination in algebra to uncertainty remaining after observation.

10.4 Quadratic forms and chi-square laws

If $Z \sim N(0,1)$, the two-branch transform $Y = Z^2$ gives $f_Y(y) = \mathrm{e}^{-y/2}/\sqrt{2\pi y}$ for $y > 0$, a Gamma$(1/2, 1/2)$ law. Independent squares add, so $\sum_{i=1}^{d} Z_i^2 \sim \chi_d^2$. Therefore for a nonsingular Gaussian vector,

$$(\boldsymbol{X} - \boldsymbol{\mu})^{\top} \Sigma^{-1} (\boldsymbol{X} - \boldsymbol{\mu}) \sim \chi_d^2.$$

Worked Example: A confidence ellipse as a probability region

For a two-dimensional normal vector, the squared standardized radius is χ_2^2, with survival $\mathrm{e}^{-r/2}$. The ellipse defined by the quadratic form at most c has probability $1 - \mathrm{e}^{-c/2}$. To contain 95%, choose $c = -2\log(.05) \approx 5.9915$, not the one-dimensional normal cutoff 1.96 or its square without checking dimension.

Standard Workflow: Gaussian questions

Establish joint Gaussianity; identify means and covariances; transform linearly or construct a residual; compute conditional mean and residual variance; standardize the final scalar event. Before writing a multivariate density, check that the covariance is invertible.

10.5 Chapter Mind Map

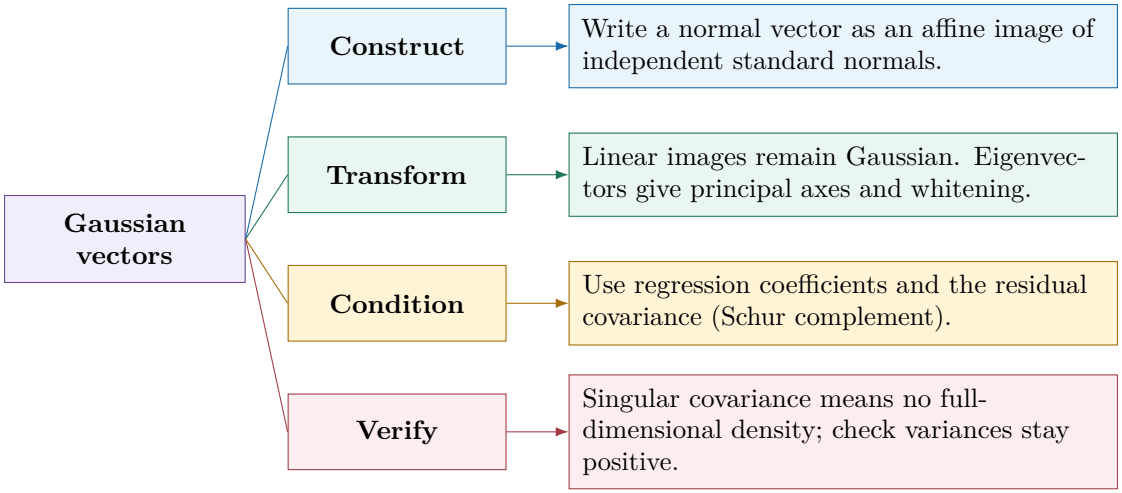


Figure 10.3: The chapter's model, calculation route, interpretation, and checks.

11 PGFs, MGFs, and Characteristic Functions

Source map: Ross 7.7; characteristic-function material explicitly supplements the syllabus's ambiguous function wording; P11.1–P11.6.

11.1 Three transforms with different domains

Definition: Choose the transform that exists

For a nonnegative integer-valued N, $G_N(s) = \mathbb{E}[s^N] = \sum_{k\geq 0} p_k s^k$, at least for $|s| \leq 1$. For real X, $M_X(t) = \mathbb{E}[\mathrm{e}^{tX}]$ is useful as an MGF when finite on an open interval containing zero. The characteristic function $\varphi_X(t) = \mathbb{E}[\mathrm{e}^{\mathrm{i}tX}]$ exists for every real t, since the integrand has modulus one.

For a PGF, $p_k = G^{(k)}(0)/k!$. Left derivatives at one give $\mathbb{E}N = G'(1-)$ and $\mathbb{E}[N(N-1)] = G''(1-)$ when finite. Thus variance is $G''(1) + G'(1) - G'(1)^2$. These are factorial moments, not the ordinary

second moment.

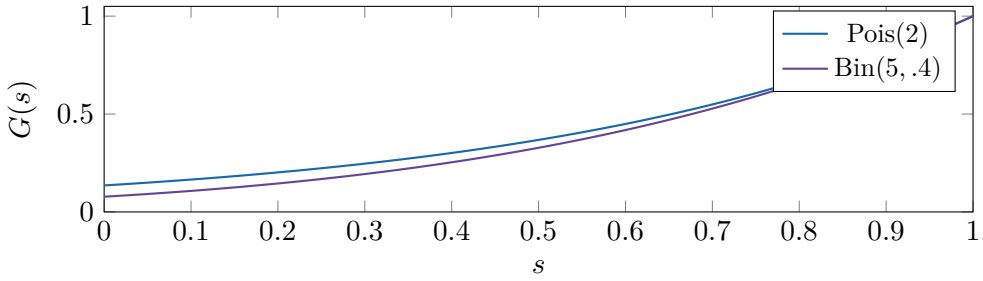


Figure 11.1: PGFs approach one at $s = 1$. Their derivatives there encode moments; their coefficients at zero encode probabilities.

Worked Example: Poisson PGF and moments

For a Poisson count, $G(s) = \mathrm{e}^{-\lambda}\sum_k (\lambda s)^k/k! = \mathrm{e}^{\lambda(s-1)}$. Hence $G'(1) = \lambda$, $G''(1) = \lambda^2$, and $\operatorname{Var} N = \lambda$. The probability of zero is $G(0) = \mathrm{e}^{-\lambda}$, not $G'(0)$.

11.2 MGFs derive moments and identify laws

If an MGF is finite near zero, exponential domination justifies differentiating under expectation there; $M^{(r)}(0) = \mathbb{E}X^r$. A distribution is uniquely determined by such an MGF. This uniqueness theorem is used here as an established transform result, not proved by matching only a finite list of moments.

For an exponential rate λ, integration gives $M(t) = \lambda/(\lambda - t)$ for $t < \lambda$. Gamma generalizes this to $(\lambda/(\lambda - t))^\alpha$. Completing the square gives the normal formula:

$$\mathrm{e}^{tx}\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)} = \mathrm{e}^{\mu t+\sigma^2 t^2/2}\mathrm{e}^{-(x-\mu-\sigma^2 t)^2/(2\sigma^2)},$$

and the remaining normal density integrates to one, so $M_X(t) = \mathrm{e}^{\mu t+\sigma^2 t^2/2}$.

Worked Example: Independent gamma sums

Independent $X \sim \text{Gamma}(a, \lambda)$ and $Y \sim \text{Gamma}(b, \lambda)$ satisfy $M_{X+Y}(t) = M_X(t)M_Y(t) = (\lambda/(\lambda - t))^{a+b}$ near zero. Uniqueness gives the gamma sum law. Different rates do not simplify to this common-rate form; the product remains a correct MGF but represents another law.

11.3 Characteristic functions: always available

Elementary properties are $\varphi(0) = 1$, $|\varphi(t)| \le 1$, $\varphi(-t) = \overline{\varphi(t)}$, and $\varphi_{aX+b}(t) = \mathrm{e}^{\mathrm{i}bt}\varphi_X(at)$. If $\mathbb{E}|X|^r < \infty$, dominated differentiation yields $\varphi^{(r)}(0) = \mathrm{i}^r\mathbb{E}X^r$. A symmetric law has a real even characteristic function.

Independence implies $\varphi_{X+Y} = \varphi_X\varphi_Y$ by factoring the expectation. The converse statement about independence requires the *joint* characteristic function to factor for all arguments: $\varphi_{X,Y}(s,t) = \varphi_X(s)\varphi_Y(t)$. Equality for the sum along $s = t$ alone does not suffice.

Key Result: Uniqueness and continuity theorems used in this course

Equal characteristic functions imply equal distributions. If $\varphi_n(t) \to \varphi(t)$ pointwise and the limit is continuous at zero, then it is a characteristic function and the corresponding laws converge in distribution. The general Fourier-analytic proofs of these theorems are outside the elementary syllabus; they are explicitly stated assumptions for the CLT derivation in Chapter 12.

Worked Example: A transform that exists without an MGF

For the symmetric two-point law $\mathbb{P}(X = \pm 1) = 1/2$, $\varphi_X(t) = \cos t$, $M_X(t) = \cosh t$. For a standard Cauchy law, the characteristic function is $e^{-|t|}$ (a standard Fourier integral), while the MGF is infinite away from zero. The cusp at zero is consistent with its undefined first moment; one must not differentiate as if it had a finite mean.

11.4 Random sums and Poisson splitting

For $S = \sum_{j=1}^{N} X_j$ with nonnegative integer N independent of iid increments, conditioning on N yields $M_S(t) = G_N(M_X(t))$ wherever finite; characteristic functions give $\varphi_S(t) = G_N(\varphi_X(t))$ without an MGF assumption. If $N \sim \text{Pois}(\lambda)$, this becomes $\exp\{\lambda(\varphi_X(t) - 1)\}$.

Worked Example: Poisson thinning proves independence, not only marginals

Classify each of $N \sim \text{Pois}(\lambda)$ occurrences independently as type one with probability p, type two otherwise. Let counts be A, B. Given $N = n$, their joint PGF is $(ps + (1-p)t)^n$. Averaging gives

$$G_{A,B}(s,t) = \exp\{\lambda[p(s-1) + (1-p)(t-1)]\} = e^{\lambda p(s-1)} e^{\lambda(1-p)(t-1)}.$$

The two-variable factorization proves independent Poisson counts with means λp and $\lambda(1-p)$. Conditional on the total, the two counts are dependent.

Standard Workflow: Transform questions

Choose a transform with a justified domain; derive it from the law; use products only for independent sums; use composition for an independent random count; differentiate at the appropriate point; invoke a stated uniqueness theorem to identify the resulting law. Check the value at zero (MGF/CF) or one (PGF) is one.

11.5 Chapter Mind Map

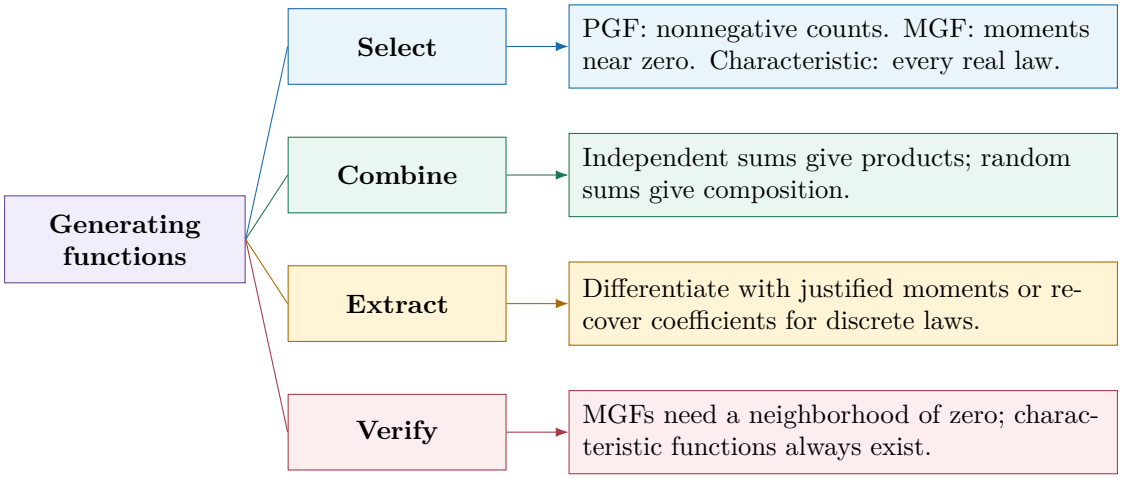


Figure 11.2: The chapter's model, calculation route, interpretation, and checks.

12 Inequalities, Laws of Large Numbers, and the CLT

Source map: Ross 8.2–8.6; P12.1–P12.6.

12.1 What kind of convergence is being claimed?

$X_n \to X$ in probability means $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$. Almost-sure convergence means the set of outcomes on which the entire sequence converges has probability one. Convergence in distribution means $F_{X_n}(x) \to F_X(x)$ at every continuity point of F_X. Almost sure convergence implies convergence in probability, which implies convergence in distribution; reverses fail in general. Convergence of CDFs alone does not guarantee convergence of moments.

12.2 Weak LLN: concentration of the sample mean

Let X_i be iid with finite mean μ and finite variance σ^2. For $\overline{X}_n = n^{-1}\sum_i X_i$, linearity and independence give $\mathbb{E}\overline{X}_n = \mu$ and $\operatorname{Var}\overline{X}_n = \sigma^2/n$. Chebyshev proves

$$\mathbb{P}(|\overline{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

The iid weak law also holds with only a finite absolute first moment, but the displayed elementary proof specifically needs finite variance. State the stronger general result separately rather than pretending this variance calculation proves it.

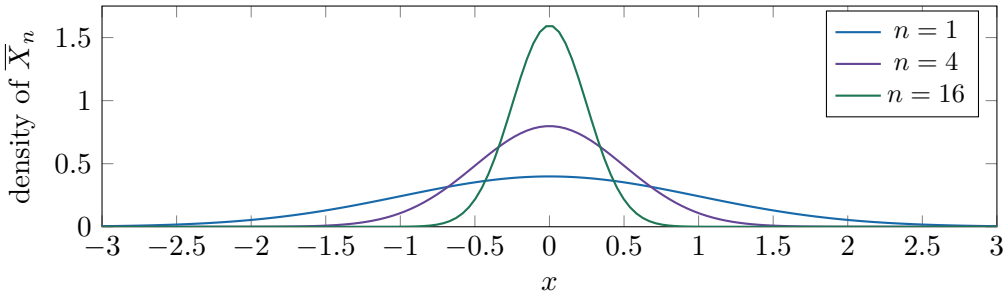


Figure 12.1: For standard normal observations these sample-mean densities are exact: their width shrinks as $1/\sqrt{n}$. The LLN describes concentration, not a fixed nonzero limiting width.

Worked Example: A guaranteed sample size

If measurements have standard deviation two and are iid, requiring $\mathbb{P}(|\overline{X}_n - \mu| \geq .1) \leq .05$ is guaranteed by $4/(.01n) \leq .05$, hence $n \geq 8000$. This is a conservative distribution-free guarantee, not the smallest possible sample size for a known normal population.

Supplementary Background: Strong LLN: statement and scope

For iid integrable variables ($\mathbb{E}|X_1| < \infty$), $\overline{X}_n \to \mu$ almost surely. A full proof requires additional convergence tools and is not claimed by the one-line Chebyshev argument. With finite variance, one can prove convergence along $n = k^2$ by summing the Chebyshev bounds and invoking Borel–Cantelli, but controlling all intermediate indices still requires another argument. This distinction prevents an incomplete proof from being presented as complete.

12.3 Central limit theorem: fluctuations on a shrinking scale

Key Result: IID finite-variance CLT

If X_i are iid, $\mathbb{E}X_i = \mu$, and $0 < \operatorname{Var} X_i = \sigma^2 < \infty$, then

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1).$$

Equivalently, $\overline{X}_n$ is approximately $N(\mu, \sigma^2/n)$ for large n. The original observations need not be normal. The theorem does not supply a universal finite-n error bound under only these assumptions.

To derive it, set $Y = (X-\mu)/\sigma$. Finite second moment and a dominated Taylor remainder give $\varphi_Y(u) = 1 - u^2/2 + o(u^2)$: the linear coefficient is $i\mathbb{E}Y = 0$, and the quadratic coefficient is $-\mathbb{E}Y^2/2 = -1/2$. One justification is that $(e^{iuY} - 1 - iuY)/u^2$ tends to $-Y^2/2$ and is bounded in modulus by $Y^2/2$. Independence now yields

$$\varphi_{Z_n}(t) = [\varphi_Y(t/\sqrt{n})]^n = \left[1 - \frac{t^2}{2n} + o(1/n)\right]^n \longrightarrow e^{-t^2/2}.$$

The characteristic-function continuity theorem stated in Chapter 11 completes the proof, identifying the standard normal limit. Thus the algebraic part is fully derived and the analytic convergence theorem used to close the argument is named explicitly.

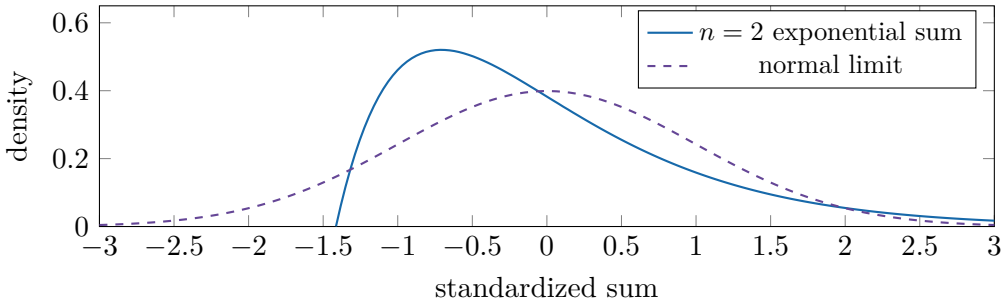


Figure 12.2: A standardized sum of two exponential waits is still skewed. The CLT predicts approach to the normal as the number of independent terms increases, not instant exact normality.

Worked Example: A sample-mean probability

For iid exponential rate two measurements, $\mu = 1/2$ and $\sigma = 1/2$. With $n = 100$, the CLT gives

$$\mathbb{P}(.45 < \overline{X}_{100} < .60) \approx \Phi(2) - \Phi(-1) \approx .8186.$$

The exact sum is Gamma$(100, 2)$, so this can also be evaluated exactly through that CDF. The approximation's standard deviation is .05, not .5 or .005.

12.4 Discrete continuity correction and approximation selection

For $B \sim \text{Bin}(n,p)$, approximate by a normal with mean np and variance $np(1-p)$ when both success and failure counts are sufficiently large for the desired accuracy. For integers $a \le b$,

$$\mathbb{P}(a \le B \le b) \approx \Phi\left(\frac{b + .5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{a - .5 - np}{\sqrt{np(1-p)}}\right).$$

For $\mathbb{P}(B \ge k)$, the lower continuous boundary is $k - .5$. The half-unit shift matches an integer mass to a unit-width interval; it is not applied to genuinely continuous sums.

Worked Example: Two suitable approximations, two regimes

For $B \sim \text{Bin}(200, .4)$, $\mathbb{P}(B \geq 90) \approx 1 - \Phi((89.5 - 80)/\sqrt{48})$. For $C \sim \text{Bin}(500, .002)$, rare events suggest Pois(1), giving $\mathbb{P}(C \geq 2) \approx 1 - 2/\text{e}$. A normal with mean one is poorly suited to this strongly skewed small-count situation. Exact binomial sums remain available in both cases.

12.5 Exponential bounds and failure checks

Markov applied to e^{tS} for $t > 0$ gives the Chernoff bound $\mathbb{P}(S \geq a) \leq \inf_{t>0} \text{e}^{-ta} M_S(t)$ where the MGF is finite. For independent Bernoulli sums, optimizing yields $\mathbb{P}(S \geq nq) \leq \text{e}^{-nD(q\|p)}$ for $p < q < 1$, with $D(q\|p) = q \log(q/p) + (1-q) \log((1-q)/(1-p))$. This is a bound, not the CLT approximation.

Worked Example: Why the assumptions are part of the answer

If every X_i is the same random variable X, then $\overline{X}_n = X$: the variance does not shrink. If the X_i are independent standard Cauchy variables, their average is still standard Cauchy, as its characteristic function is $[\text{e}^{-|t|/n}]^n = \text{e}^{-|t|}$. The finite-mean and finite-variance hypotheses fail, so neither the usual mean LLN nor the finite-variance CLT applies.

Standard Workflow: Limit and sample-size questions

Identify sum versus mean; state iid and moment conditions; decide exact law, guaranteed bound, or approximation; compute center and scale; translate endpoints (add a continuity correction for counts); solve the probability or sample-size inequality; round sample size upward and label the conclusion honestly.

12.6 Chapter Mind Map

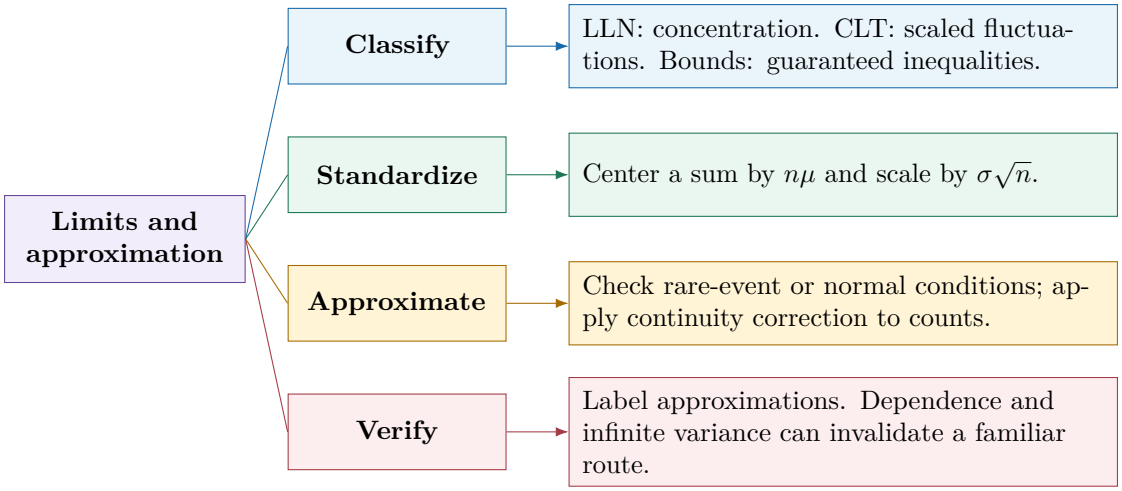


Figure 12.3: The chapter's model, calculation route, interpretation, and checks.

A Formula Index and Method Selection

A.1 Core distribution index

All parameters below use rates rather than scales. Geometric and negative binomial variables count trials.

Law	Support / parameters	Mean	Variance
Bernoulli	$0,1$; $p \in [0,1]$	p	$p(1-p)$
Binomial	$0,\dots,n$	np	$np(1-p)$
Geometric	$1,2,\dots$; $p>0$	$1/p$	$(1-p)/p^2$
Negative binomial	$r, r+1,\dots$	r/p	$r(1-p)/p^2$
Hypergeometric	population N, successes K, draws n	nK/N	$np(1-p)(N-n)/(N-1)$, $p=K/N$
Poisson	$0,1,\dots$; $\lambda>0$	λ	λ
Uniform	(a,b)	$(a+b)/2$	$(b-a)^2/12$
Exponential	$x>0$; rate λ	$1/\lambda$	$1/\lambda^2$
Gamma	$x>0$; shape α, rate λ	α/λ	α/λ^2
Beta	$(0,1)$; $\alpha,\beta>0$	$\alpha/(\alpha+\beta)$	$\alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)]$
Normal	$\mathbb{R}$; $\mu, \sigma^2>0$	μ	σ^2
Chi-square	$x>0$; ν df	ν	2ν

A.2 Short transform index

$$G_{\mathrm{Bin}(n,p)}(s) = (1-p+ps)^n, \qquad G_{\mathrm{Pois}(\lambda)}(s) = \mathrm{e}^{\lambda(s-1)},$$
$$M_{N(\mu,\sigma^2)}(t) = \mathrm{e}^{\mu t+\sigma^2 t^2/2}, \qquad M_{\mathrm{Gamma}(a,\lambda)}(t) = (\lambda/(\lambda-t))^a,\ t<\lambda,$$
$$\varphi_{N(\mu,\sigma^2)}(t) = \mathrm{e}^{\mathrm{i}\mu t-\sigma^2 t^2/2}, \qquad \varphi_{aX+b}(t) = \mathrm{e}^{\mathrm{i}bt}\varphi_X(at).$$

A.3 Six recurring decisions

1. Event wording: translate "at least," "given," "exactly," and "first" before calculating.
2. Requested law: retain support, atoms, inverse branches, and normalization.
3. Requested moment: try LOTUS, indicators, tails, or conditioning before transforming a density.
4. Joint law: draw support before selecting integration limits; do not infer independence from marginals.
5. Normal vector: use affine maps and residuals, with covariance invertibility checked where needed.
6. Asymptotics: distinguish a theorem about a limit, a finite-sample guarantee, and a numerical approximation.

A.4 Source and completeness record

The supplied syllabus image and the supplied 484-page ninth-edition Ross PDF were inspected. Subject coverage is organized around Ross Chapters 1–8, with textbook section references at chapter openings and a syllabus map in the front matter. Mixed-law bookkeeping, characteristic functions, and vector Gaussian derivations are explicitly expanded. Source textbook exercises inform mathematical scope,

but all practice questions and instructional prose here are newly written. No claims of school-specific exam frequencies or exhaustive prediction of future question variants are made. The matching practice volume supplies six worked multipart problems per learning chapter.

A.5 Frequently Needed Integrals and Series

Each identity below includes the domain condition that makes it usable. They reduce normalization, moments, and generating-function calculations to recognizable forms.

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}, \qquad \sum_{k=1}^{\infty} k q^{k-1} = \frac{1}{(1-q)^2}, \qquad |q| < 1,$$

$$\sum_{k=2}^{\infty} k(k-1) q^{k-2} = \frac{2}{(1-q)^3}, \qquad \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = \mathrm{e}^{\lambda},$$

$$\int_0^{\infty} x^{a-1} \mathrm{e}^{-\lambda x} \mathrm{d}x = \frac{\Gamma(a)}{\lambda^a}, \qquad \int_0^1 x^{a-1}(1-x)^{b-1} \mathrm{d}x = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

$$\int_{\mathbb{R}} \mathrm{e}^{-ax^2} \mathrm{d}x = \sqrt{\frac{\pi}{a}}, \qquad \int_0^1 x^r (-\log x) \mathrm{d}x = \frac{1}{(r+1)^2}.$$

The gamma and beta integrals require $a, b, \lambda > 0$; the Gaussian integral requires $a > 0$; the logarithmic integral requires $r > -1$. The last identity follows by differentiating $\int_0^1 x^r \mathrm{d}x = 1/(r+1)$ with respect to r and changing sign. The normal even moments satisfy $\mathbb{E}Z^{2m} = (2m-1)\mathbb{E}Z^{2m-2}$ by integration by parts, starting from $\mathbb{E}Z^0 = 1$. Thus $\mathbb{E}Z^4 = 3$ and $\mathbb{E}Z^6 = 15$; odd moments vanish by symmetry.

A.6 Stable Calculations and Finite Uniform Laws

For $X \sim \mathrm{Bin}(n, p)$, with $0 < p < 1$, adjacent masses obey

$$\frac{p_X(k+1)}{p_X(k)} = \frac{n-k}{k+1} \frac{p}{1-p}.$$

For a Poisson law the ratio is $\lambda/(k+1)$. These identities follow by cancelling factorials and are useful for summing tails without repeatedly evaluating large factorials. Handle $p = 0$ and $p = 1$ as degenerate binomial laws rather than dividing by zero. Complementing an extreme tail can lose numerical precision; sum the small tail directly when high accuracy is needed.

For completeness, the discrete uniform law on $\{1, \ldots, m\}$ assigns mass $1/m$ at each point. The identities $\sum k = m(m+1)/2$ and $\sum k^2 = m(m+1)(2m+1)/6$ give mean $(m+1)/2$ and variance $(m^2-1)/12$. An affine transformation produces a uniform law on an equally spaced finite grid, not a continuous uniform density. For a fair die, mean $7/2$ and variance $35/12$ follow at $m = 6$.